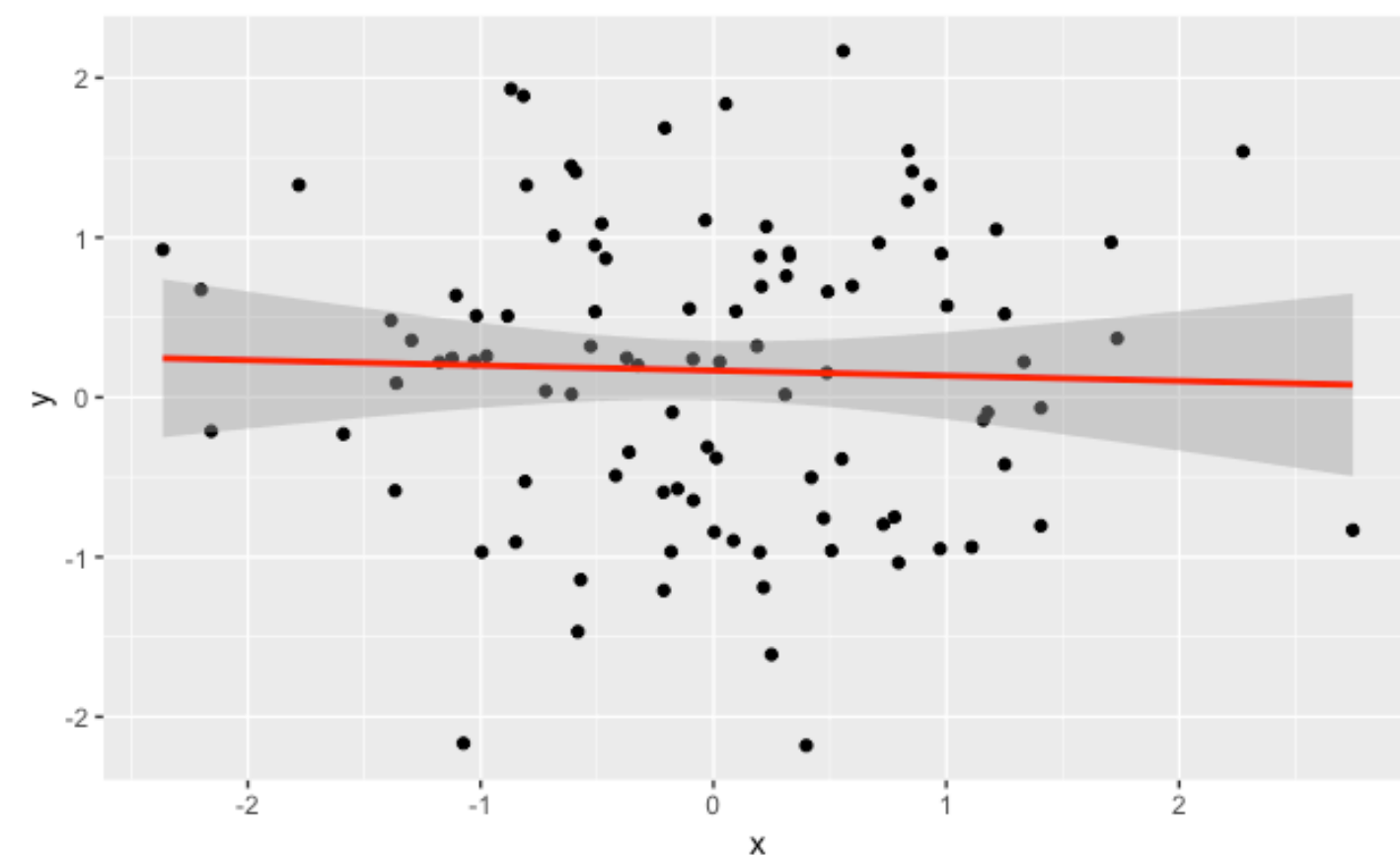
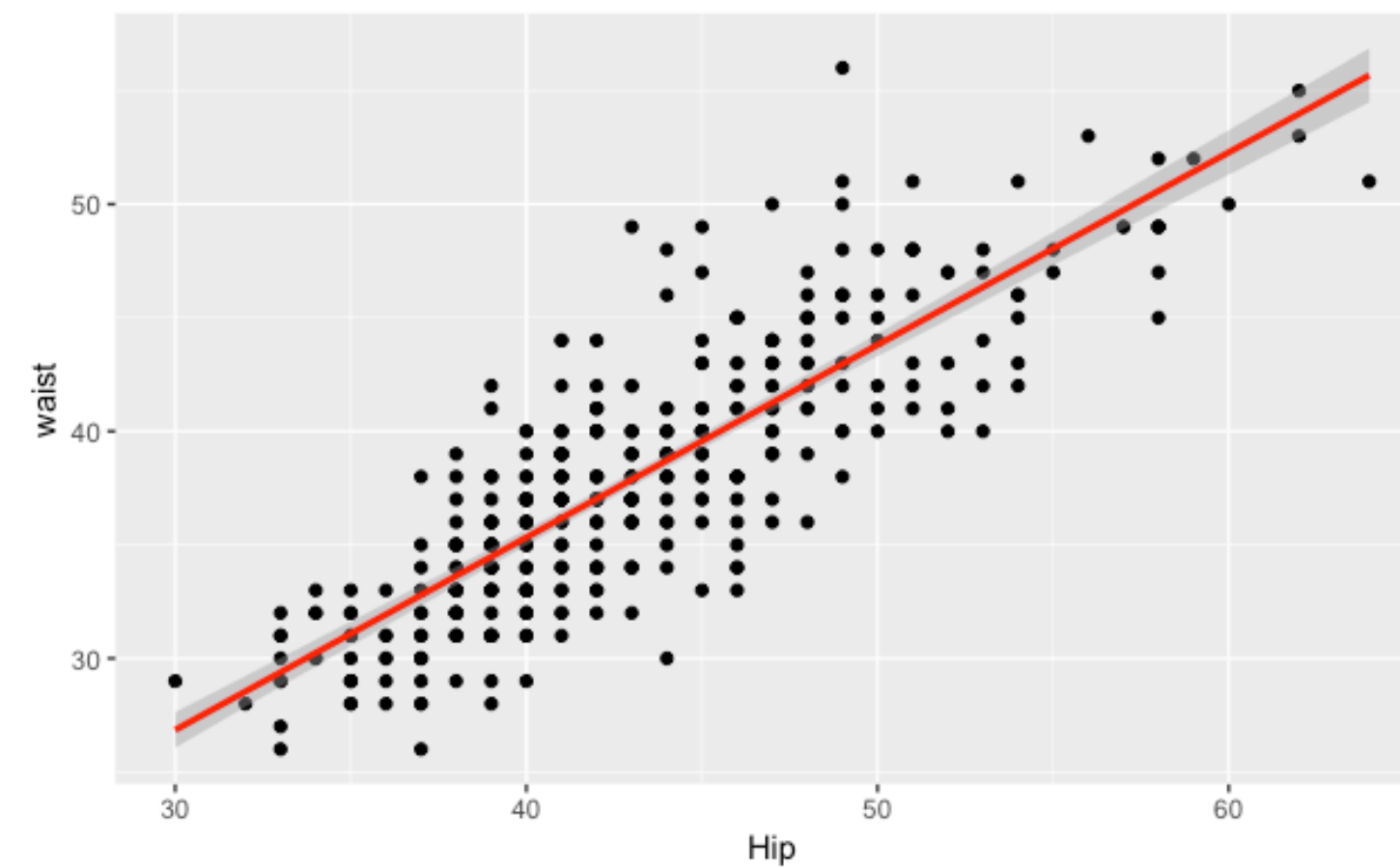
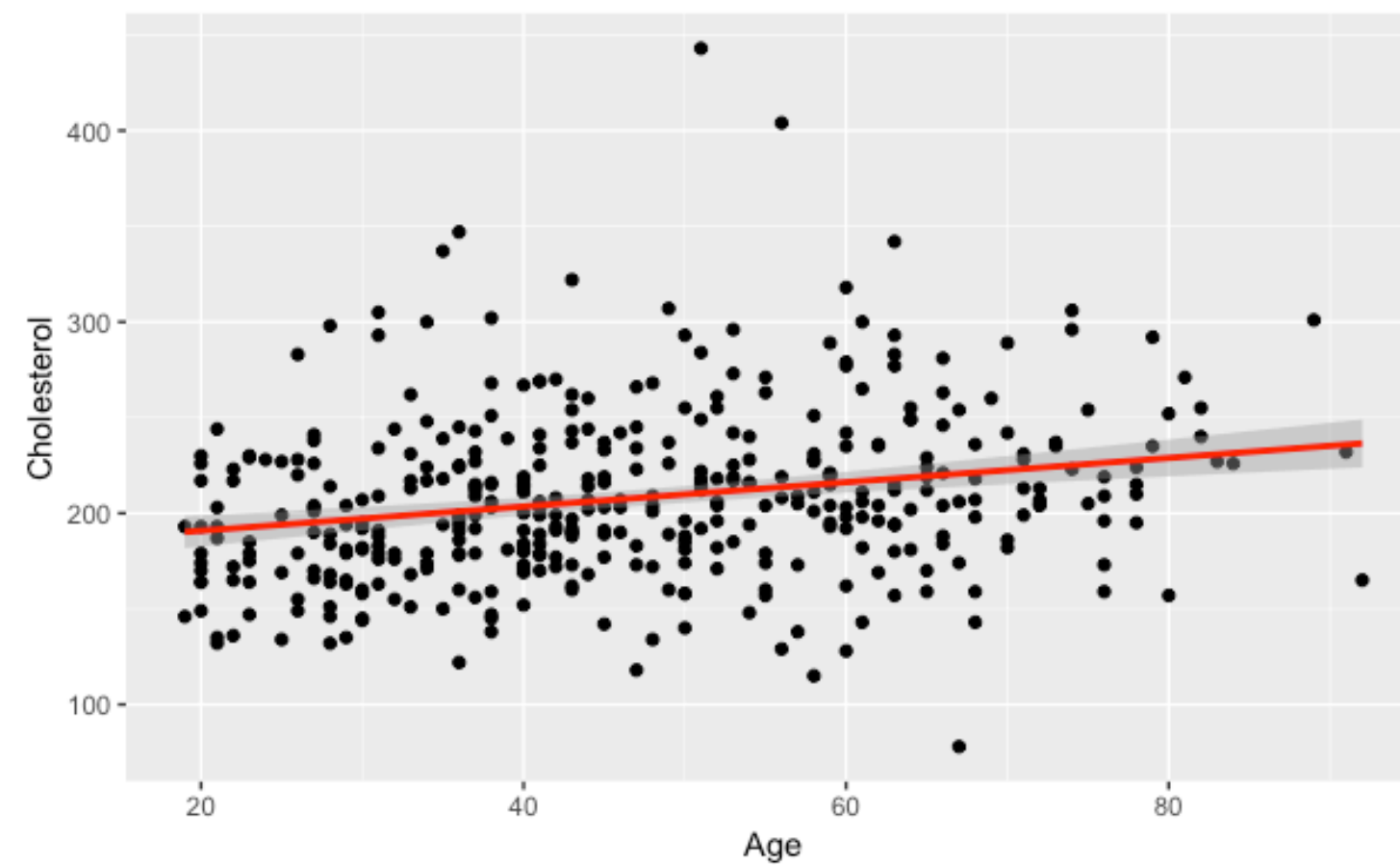


2. Describing data - correlations

Relation between numerical variables

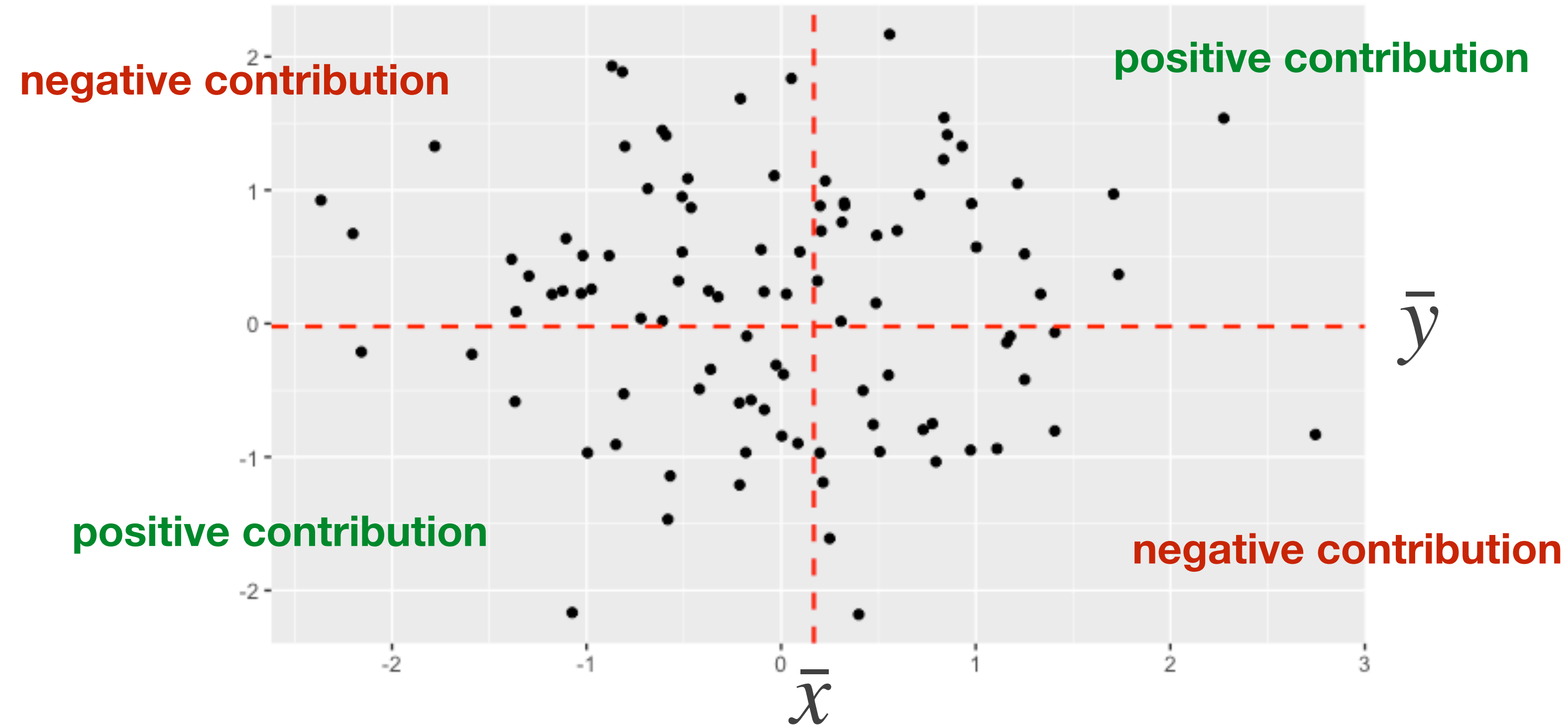
- How easy is it to draw a line through a scatter plot?



Relation between numerical variables

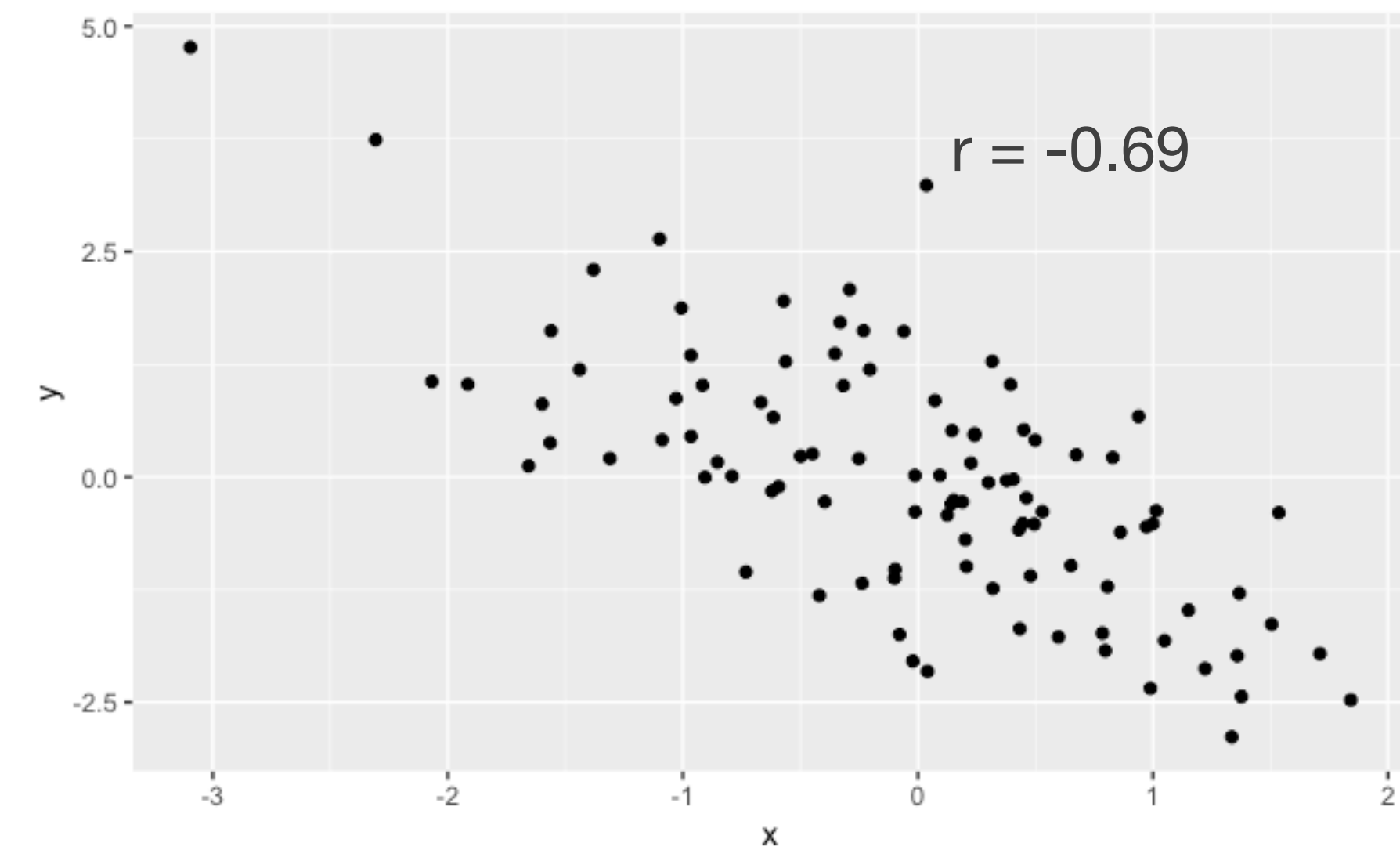
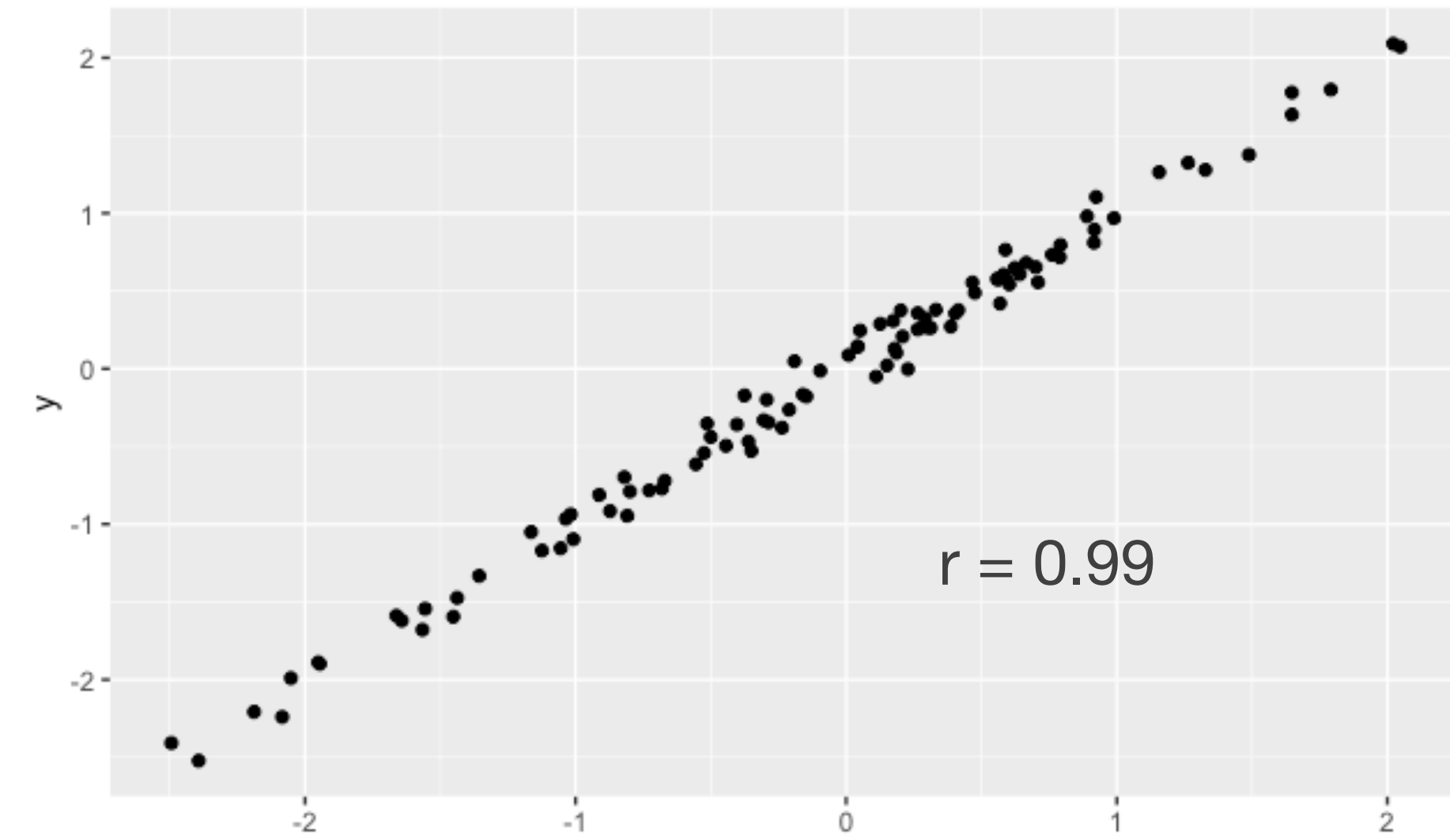
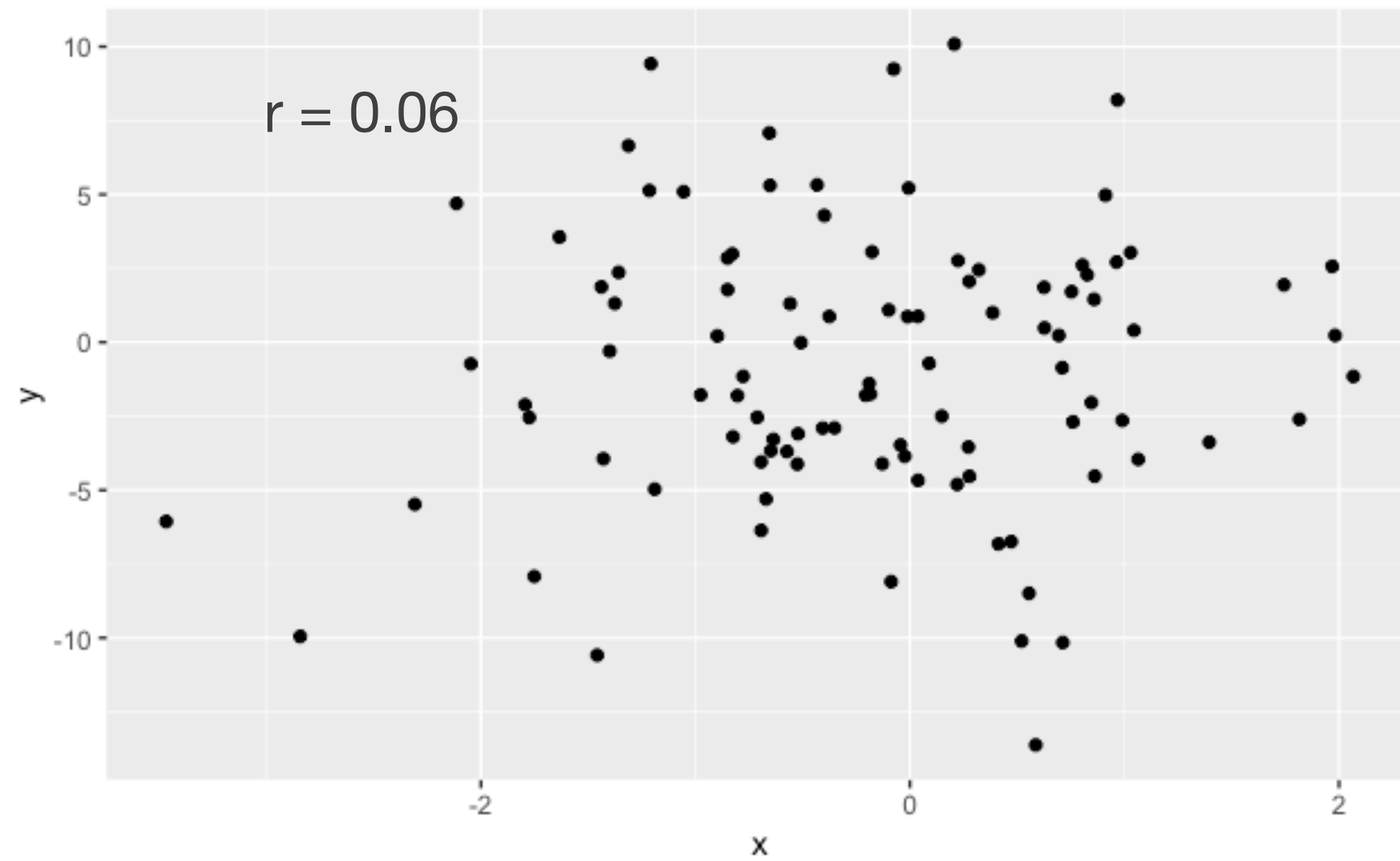
- Variance:
$$Var(x) = (s_x)^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$
 dimension: $[x]^2$
- Covariance :
$$Cov(x, y) = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$
 dimension: $[x][y]$
- Pearson Correlation :
$$Corr(x, y) = r = \frac{1}{N-1} \sum_{i=1}^N \frac{(x_i - \bar{x})}{s_x} \frac{(y_i - \bar{y})}{s_y}$$
 dimension: none
- Properties:
 - correlation is scale invariant, covariance is not!
 - $cor(x, x) = 1$
 - $-1 \leq cor(x, y) \leq +1$

Relation between numerical variables



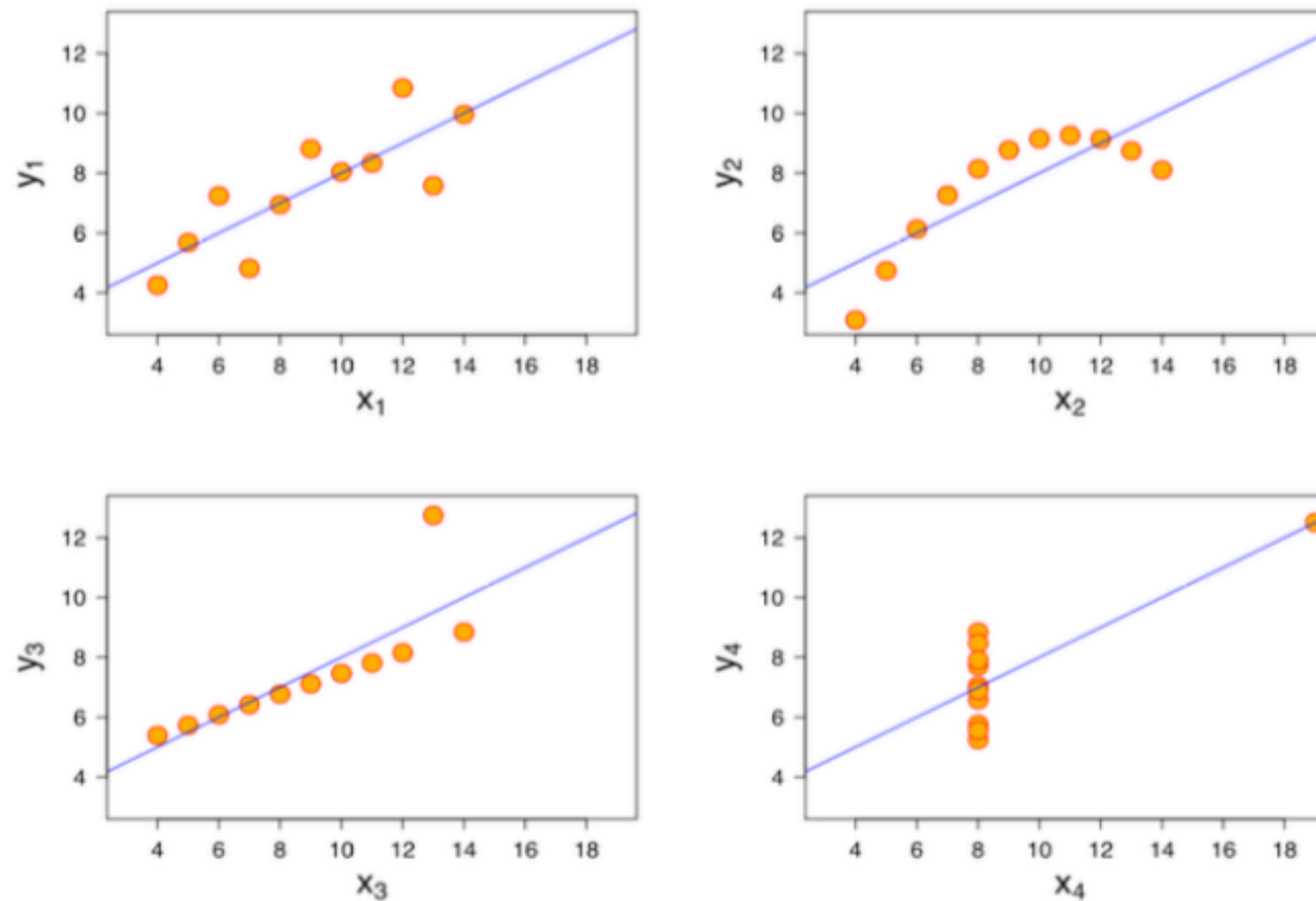
$$\text{Corr}(x, y) = \frac{1}{N-1} \sum_{i=1}^N \frac{(x_i - \bar{x})}{s_x} \frac{(y_i - \bar{y})}{s_y}$$

Relation between numerical variables



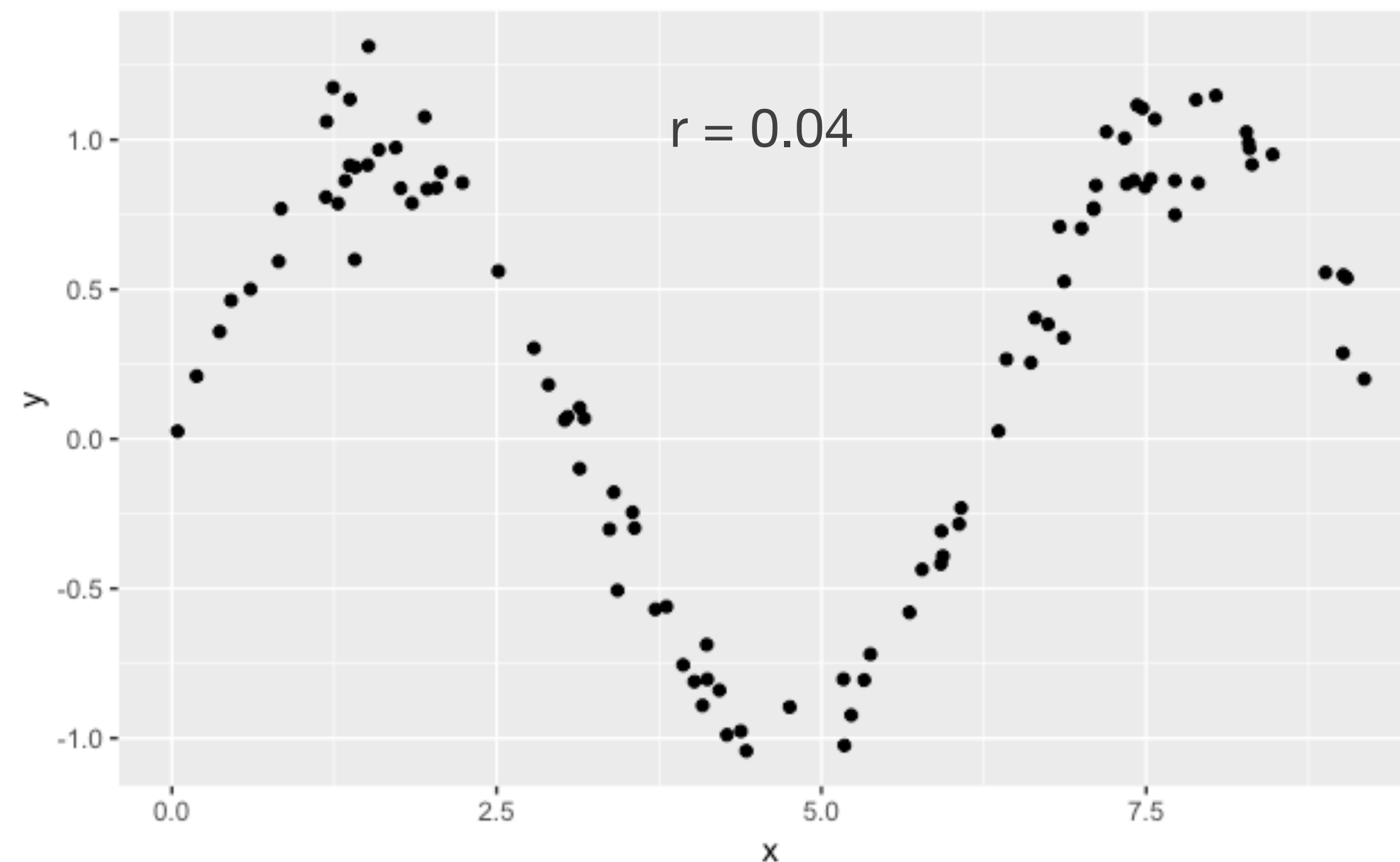
Relation between numerical variables

- **Anscombe quartet** is a dataset to highlight possible caveats of correlation measures
- Correlation is equal in all 4 cases ($r = 0.82$)



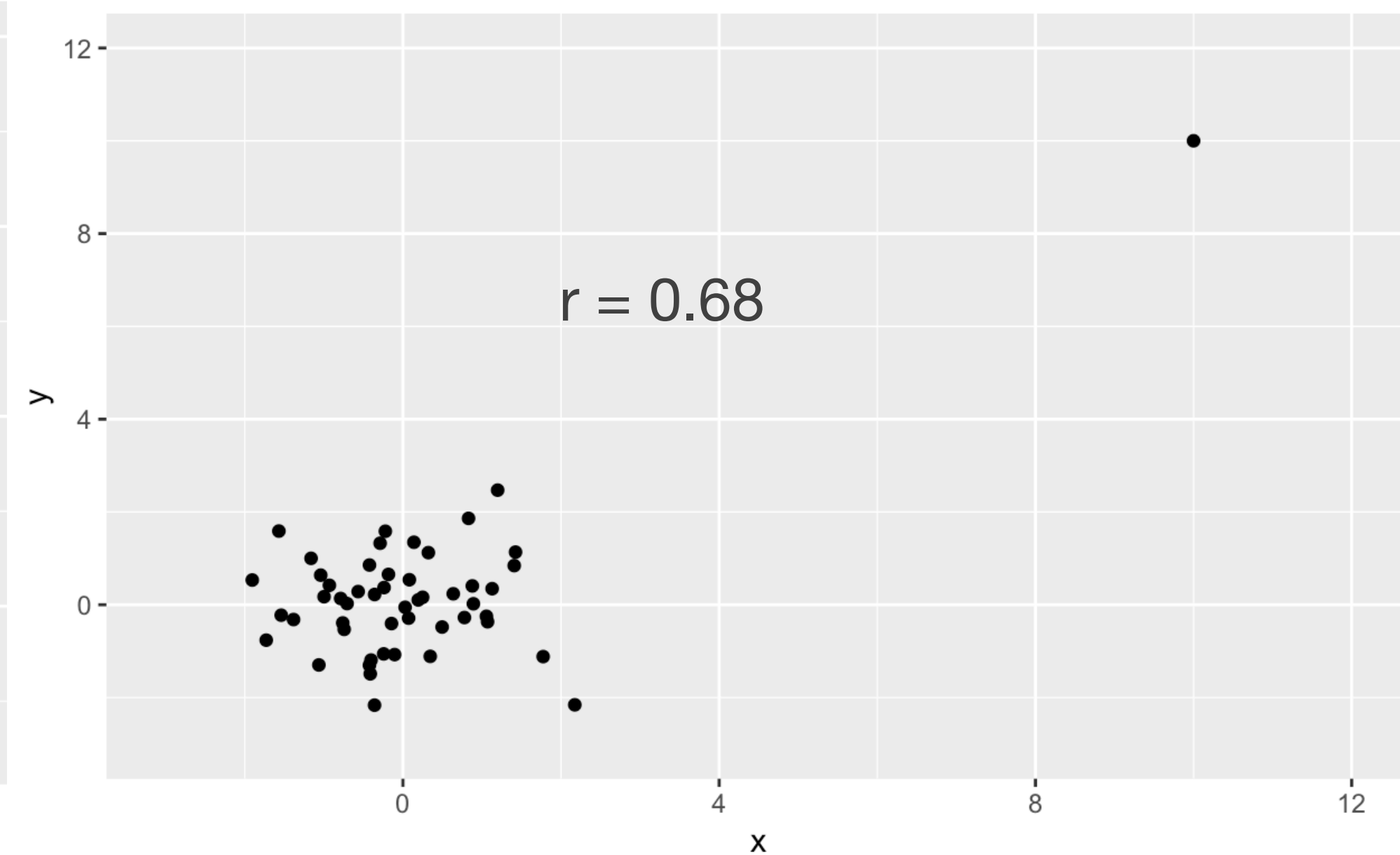
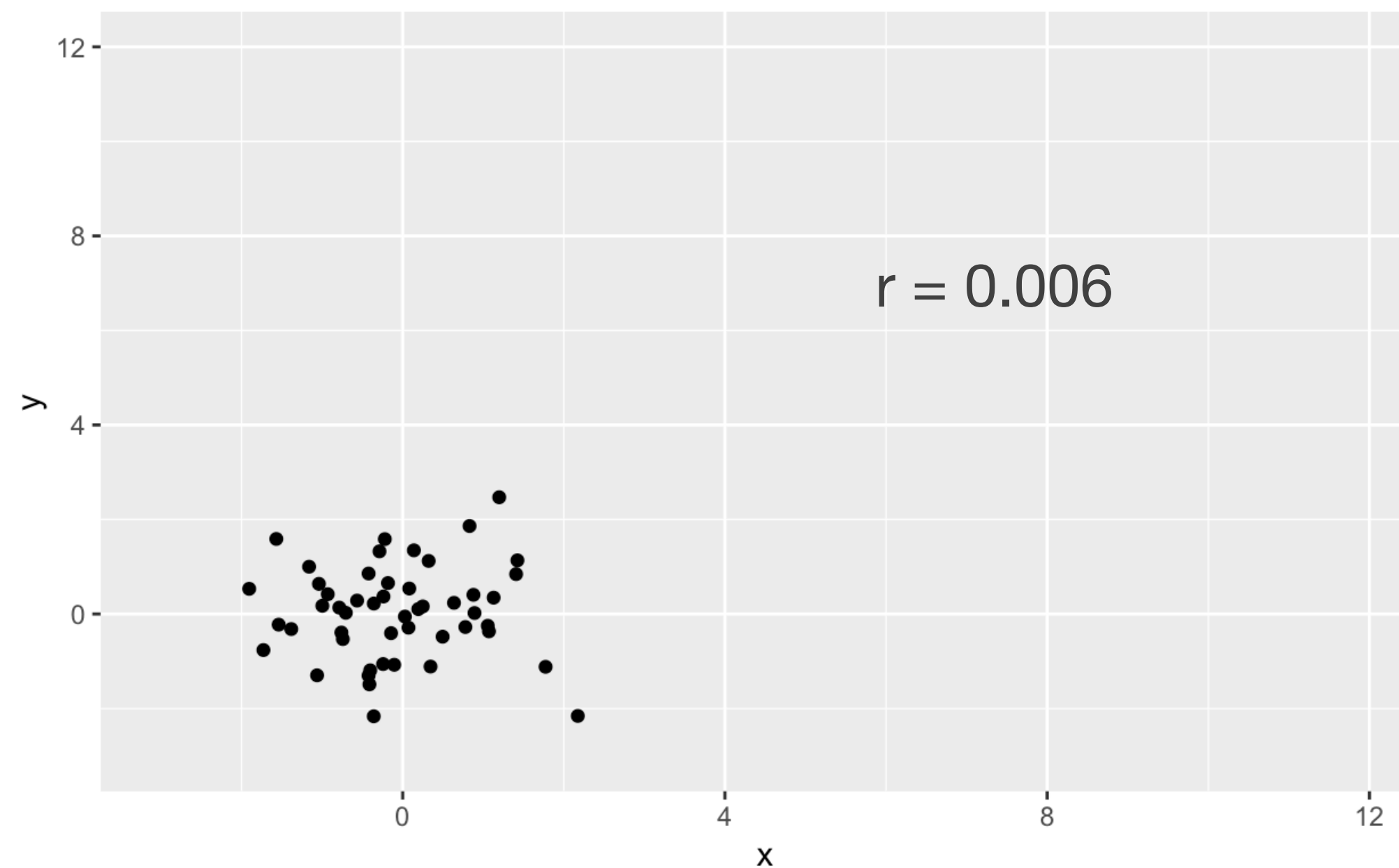
Relation between numerical variables

- Beware that correlation estimates a **linear** relationship!
- Weak correlation **does NOT mean** that there is no relation between both variables!



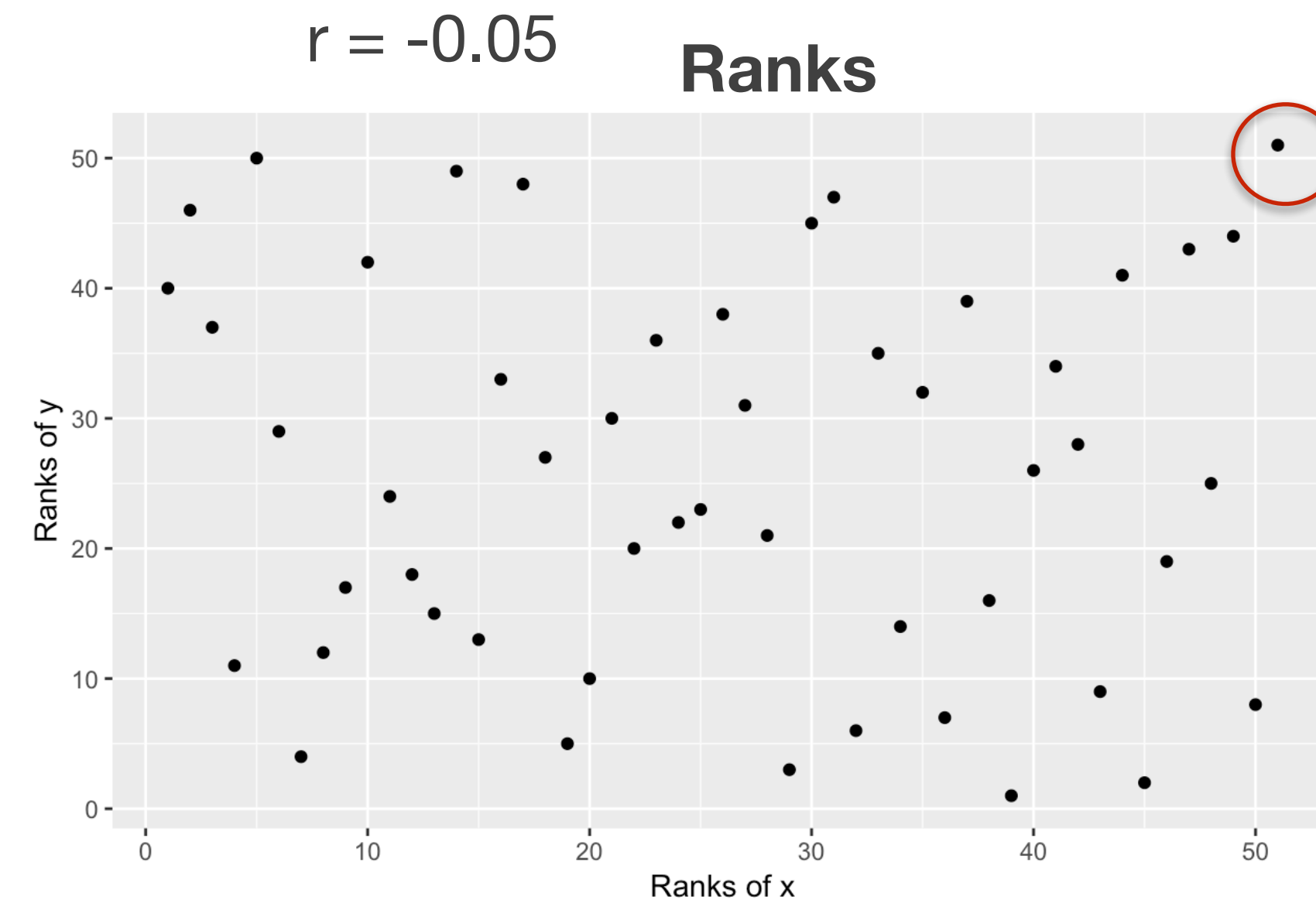
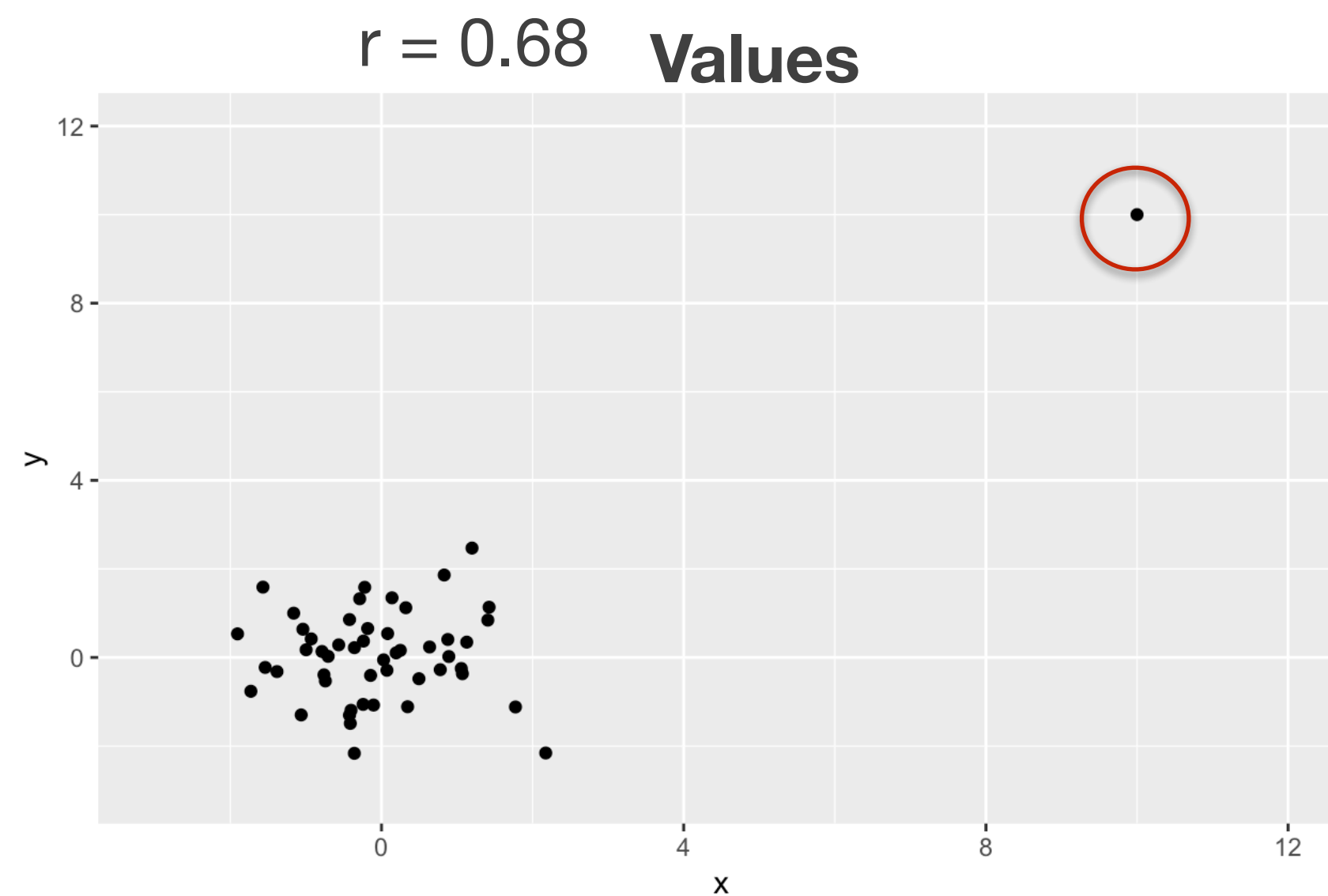
Correlation and outliers

- Pearson correlation is strongly influenced by outliers!
- we need to find robust measure which is insensitive to outliers.

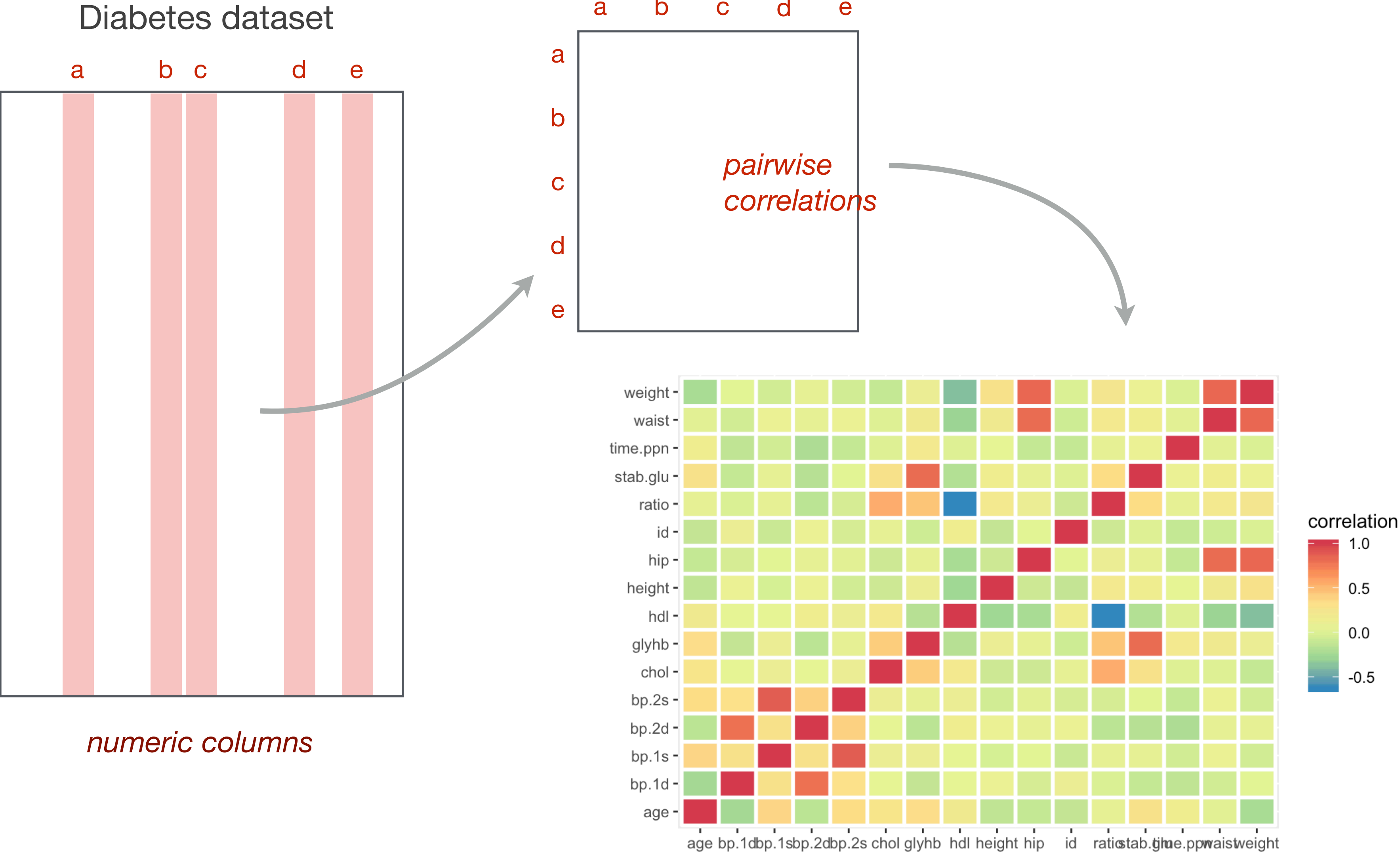


Correlation and outliers

- Transform values to ranks
 $x = \{2.3, -1.7, 1.1, -0.6, 12.3\} \rightarrow R(x) = \{4, 1, 3, 2, 5\}$
 $y = \{-2.7, 0.8, 3.1, -1.7, 14.3\} \rightarrow R(y) = \{1, 3, 4, 2, 5\}$
- Compute the correlation **on the ranks** : $\text{cor}(R(x), R(y))$
= **Spearman correlation**



Correlating multiple variables



3. Cleaning data

Dealing with weird values

- Datasets originate from measurements, and might contain **unexpected values**
 - missing values
 - outliers
- Examples:
 - cholesterol measurement for one patient was not made and is missing
→ how is this indicated in the dataset? 0 or “NA” (not available)
 - machine was wrongly calibrated for one experiment and returned an unusually high value
→ technical artifact or biologically relevant phenomenon?
- Outliers might have a **technical** origin (artifacts) or a **biological** origin (interesting!)

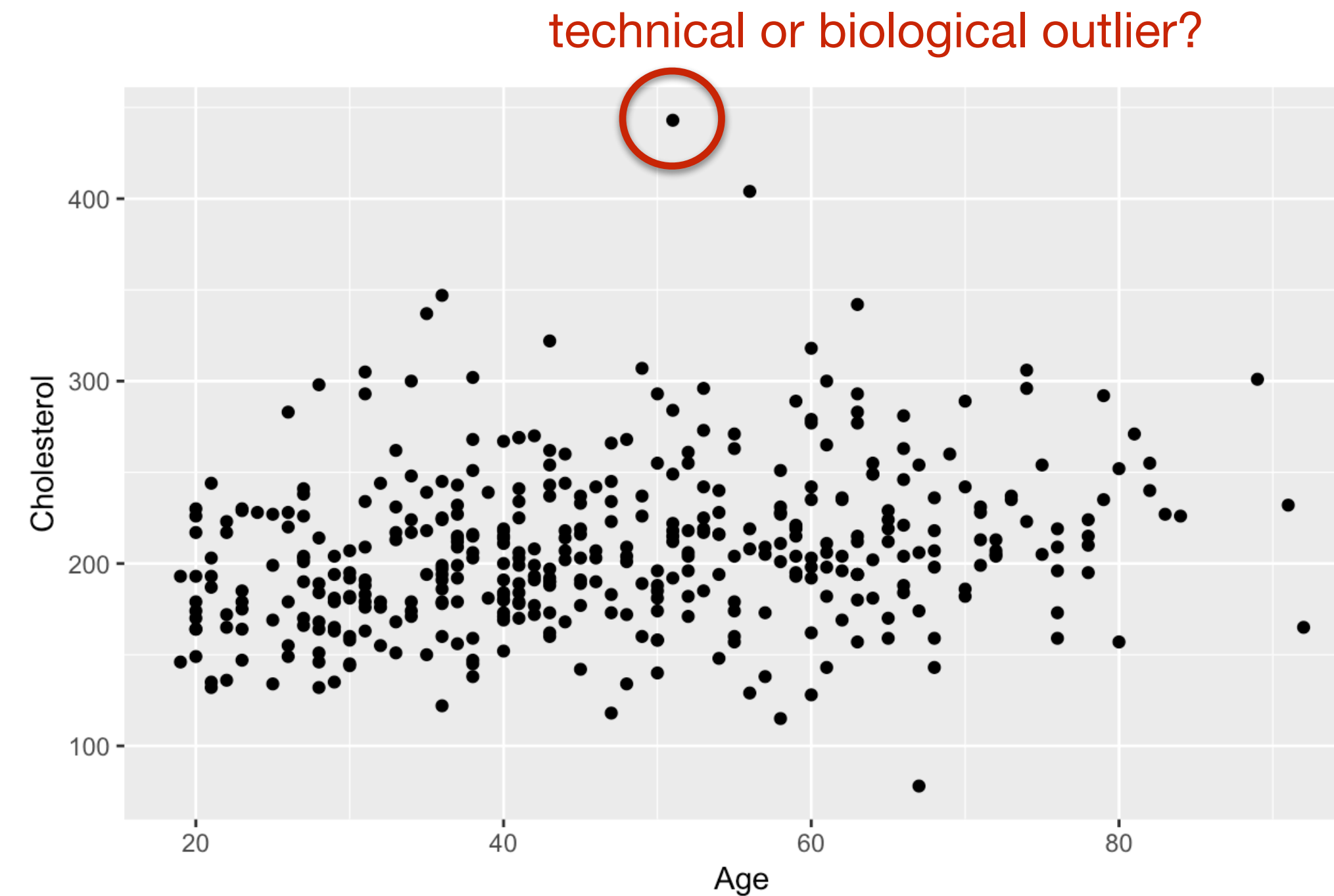
Dealing with weird values

- Solutions:
 - remove rows / columns containing NA values
 - replace missing values with “likely” values (e.g. mean of other available values)
→ **imputation**
- Possible imputation strategies
 - replace missing value with **mean** of existing values
 - draw **random value** from theoretical distribution
 - **predict** missing values from other variables

missing NA values

id	height	weight	frame	bp.1s	bp.1d	bp.2s	bp.2d	waist	hip	time.ppn
1000	62	121	medium	118	59	NA	NA	29	38	720
1001	64	218	large	112	68	NA	NA	46	48	360
1002	61	256	large	190	92	185	92	49	57	180
1003	67	119	large	110	50	NA	NA	33	38	480
1005	68	183	medium	138	80	NA	NA	44	41	300
1008	71	190	large	132	86	NA	NA	36	42	195
1011	69	191	medium	161	112	161	112	46	49	720

Dealing with weird values

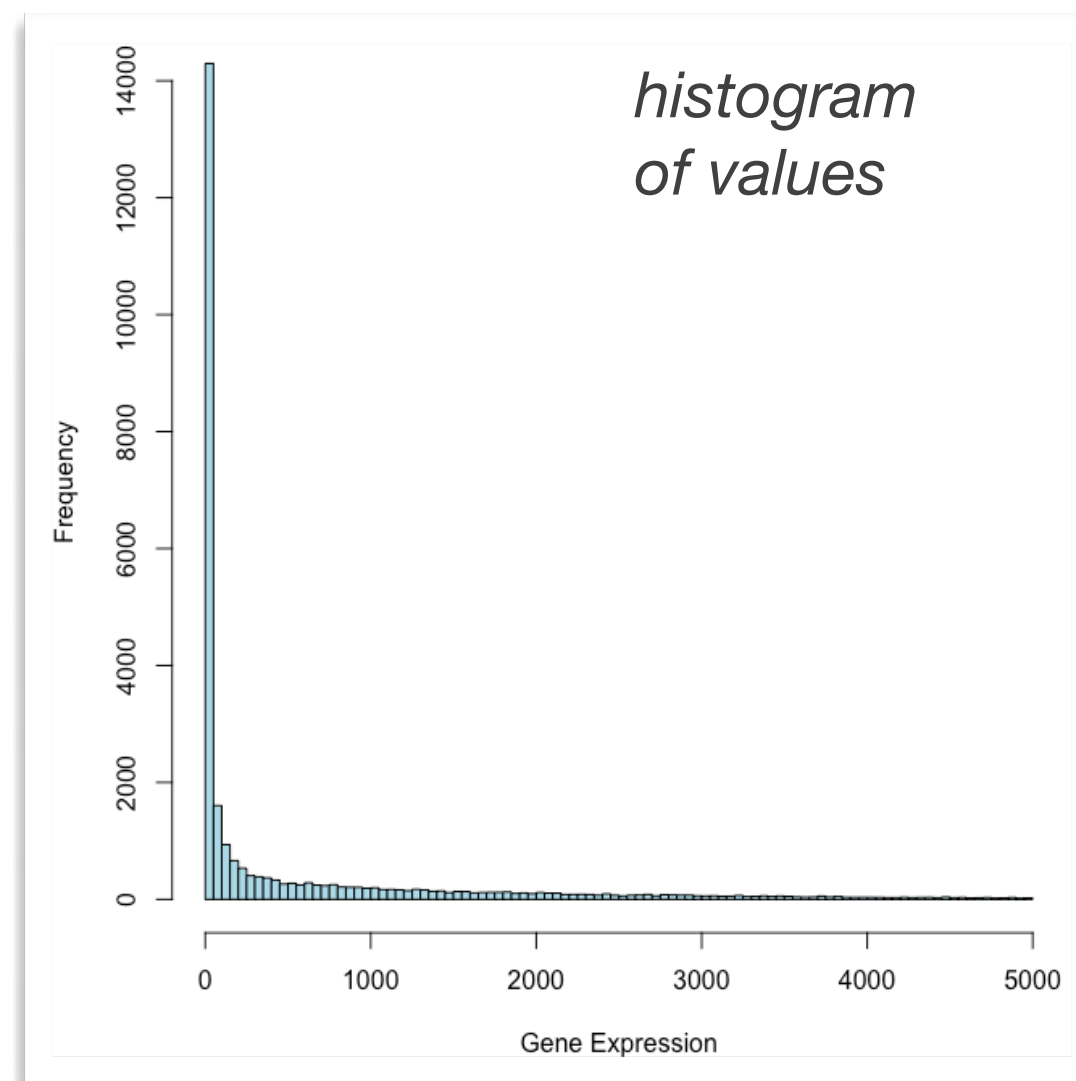


- Solutions:
 - if artifact, use measures that are less sensitive to outliers
 - **remove** outlier values from dataset
 - **replace** outlier value with more likely value (see imputation strategies)

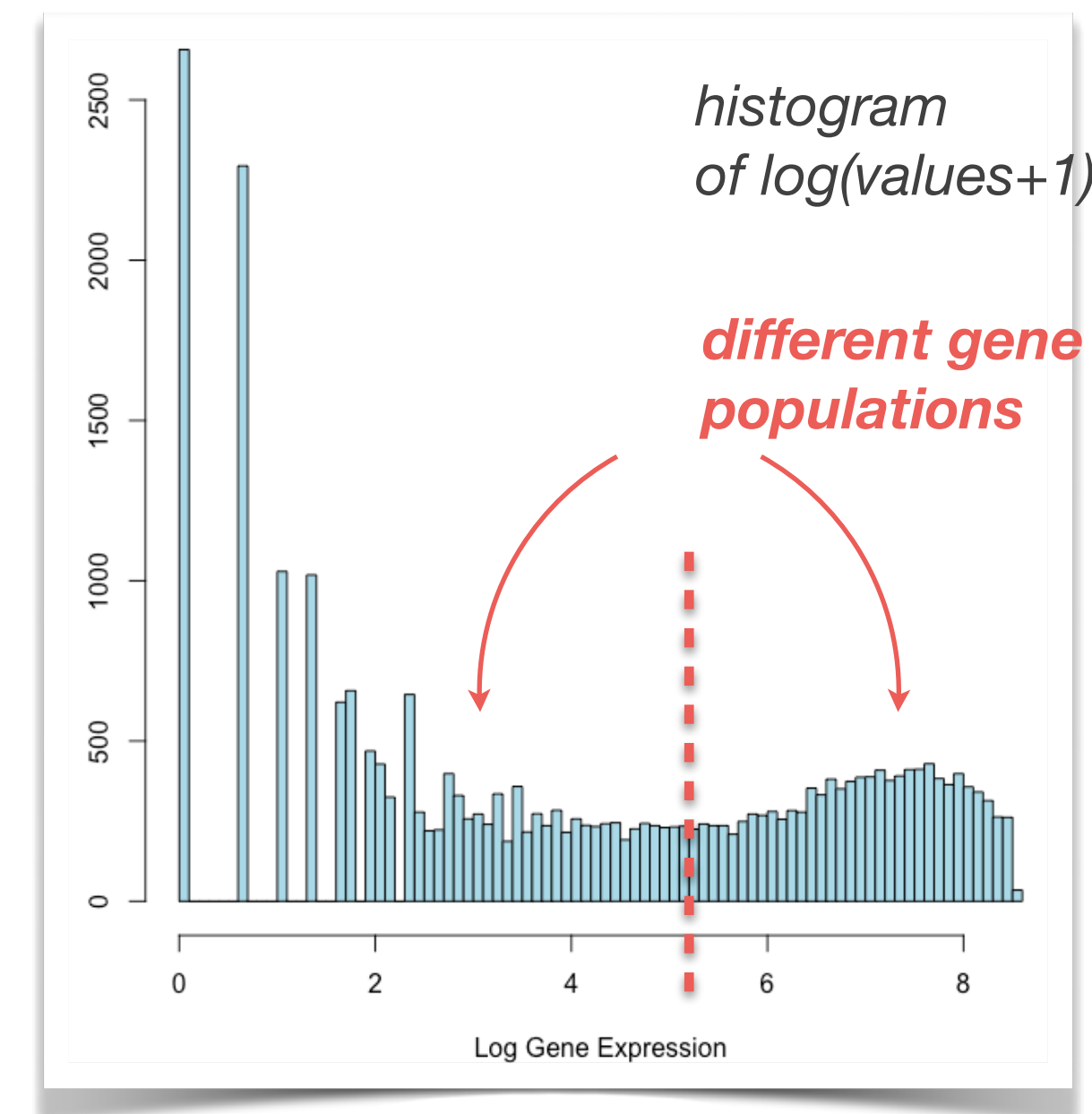
Dealing with skewed data

- Many data type have **heavily skewed distributions**

NA06985	NA06986	NA06994	NA07000	NA07037	NA07051
Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.0	Min. : 0.00	Min. : 0.00
1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.0	1st Qu.: 0.00	1st Qu.: 0.00
Median : 0.00	Median : 0.00	Median : 0.00	Median : 0.0	Median : 0.00	Median : 0.00
Mean : 21.51	Mean : 22.33	Mean : 16.96	Mean : 20.6	Mean : 20.83	Mean : 20.58
3rd Qu.: 0.00	3rd Qu.: 0.00	3rd Qu.: 0.00	3rd Qu.: 0.0	3rd Qu.: 0.00	3rd Qu.: 0.00
Max. : 36671.00	Max. : 36065.00	Max. : 30852.00	Max. : 19477.0	Max. : 29761.00	Max. : 38180.00



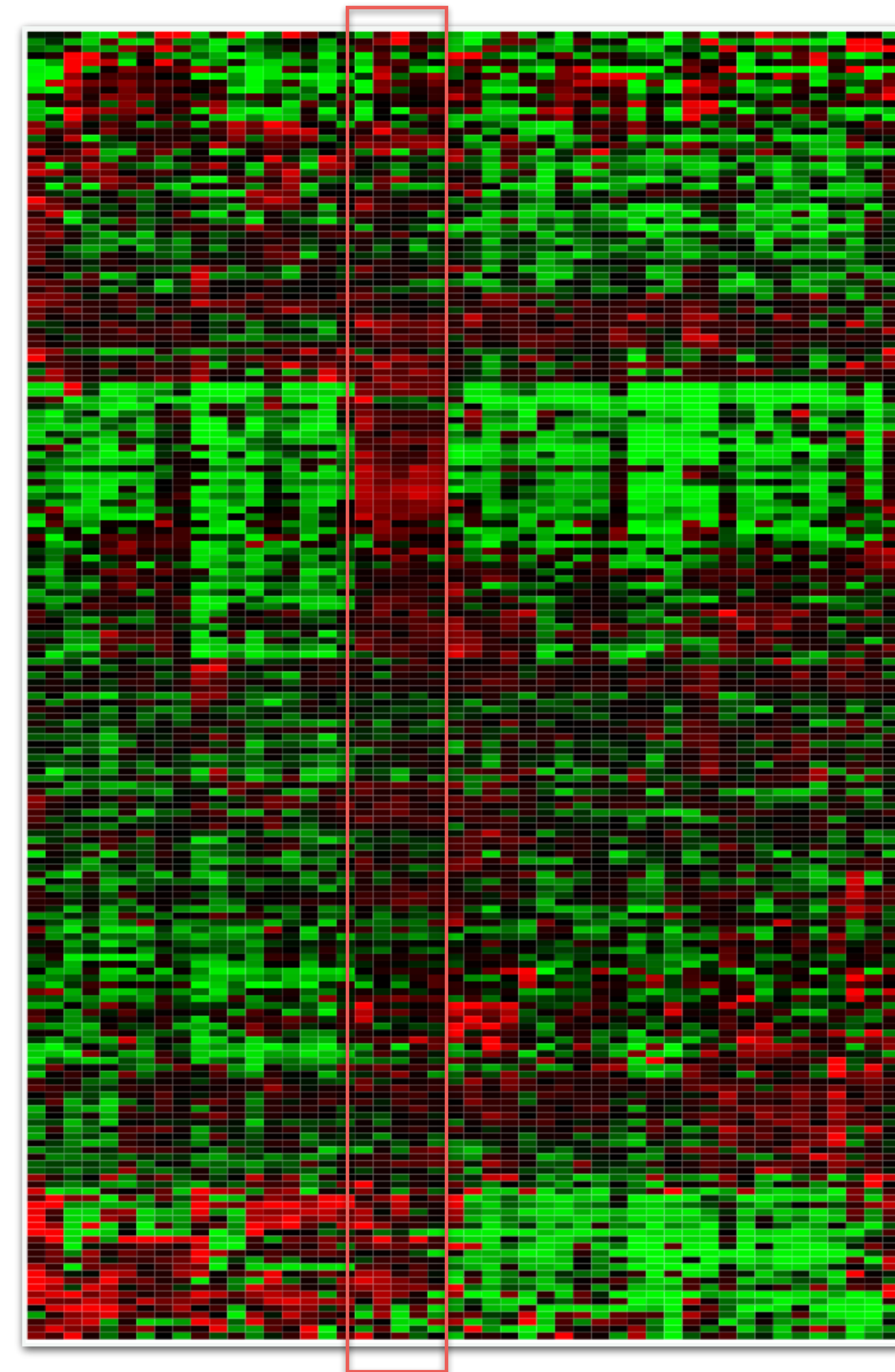
log-transformation



4. Finding structure in the data - Principles of clustering

Finding structure in the data

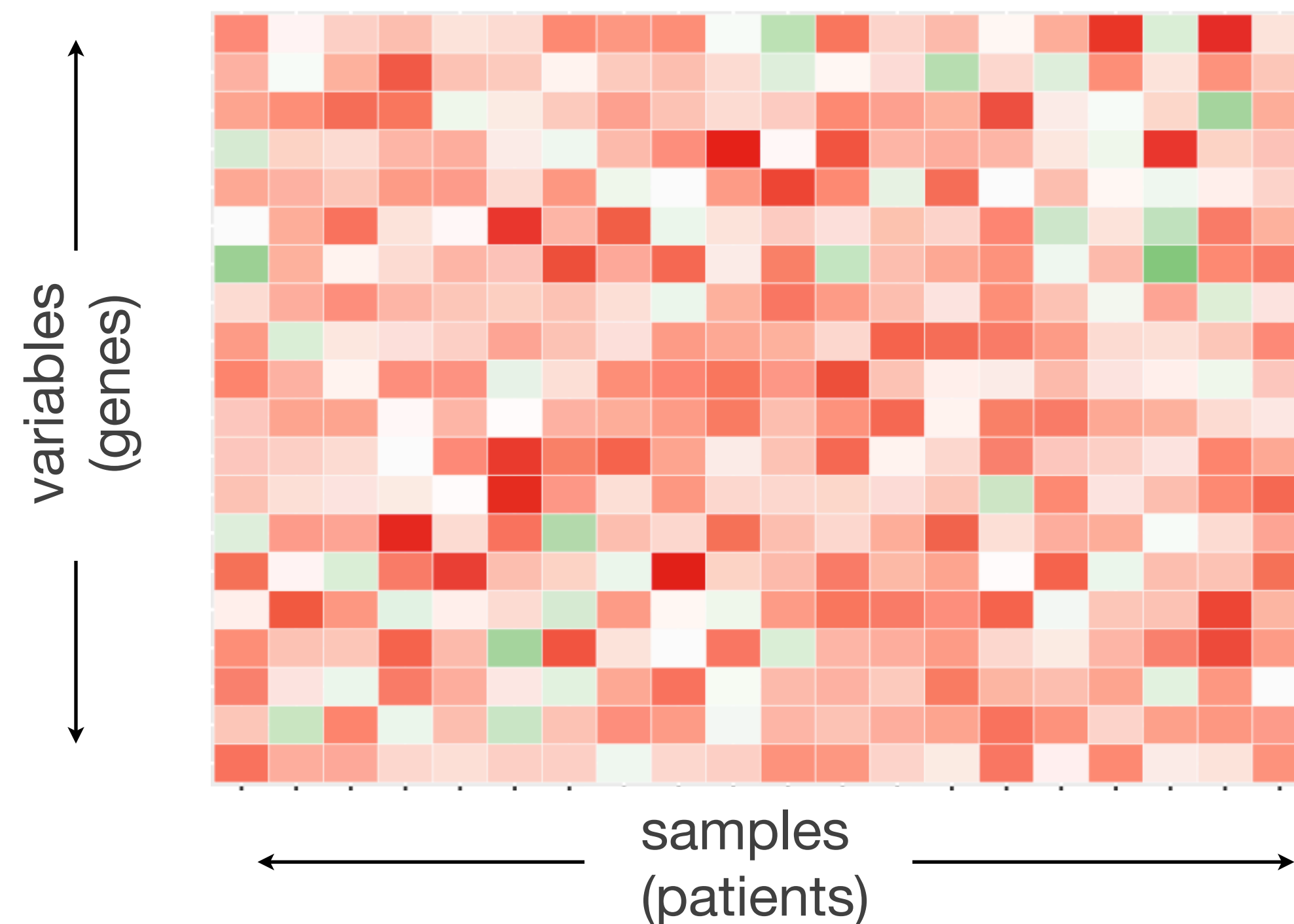
- Goal:
use a dataset to identify groups of samples
which seem to be similar
- Example
 - gene expression data
 - clinical data
 - ...
- Important for molecular subtyping

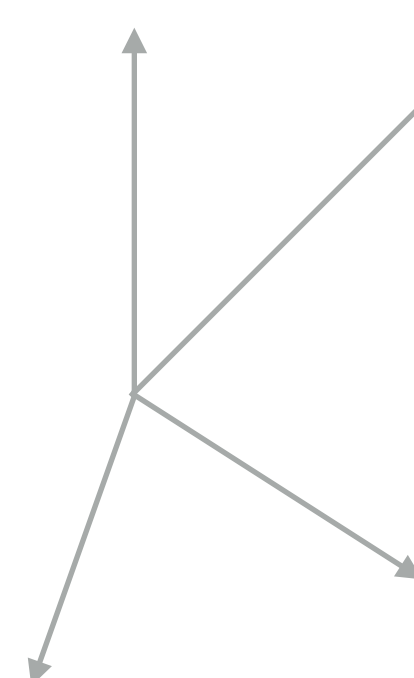
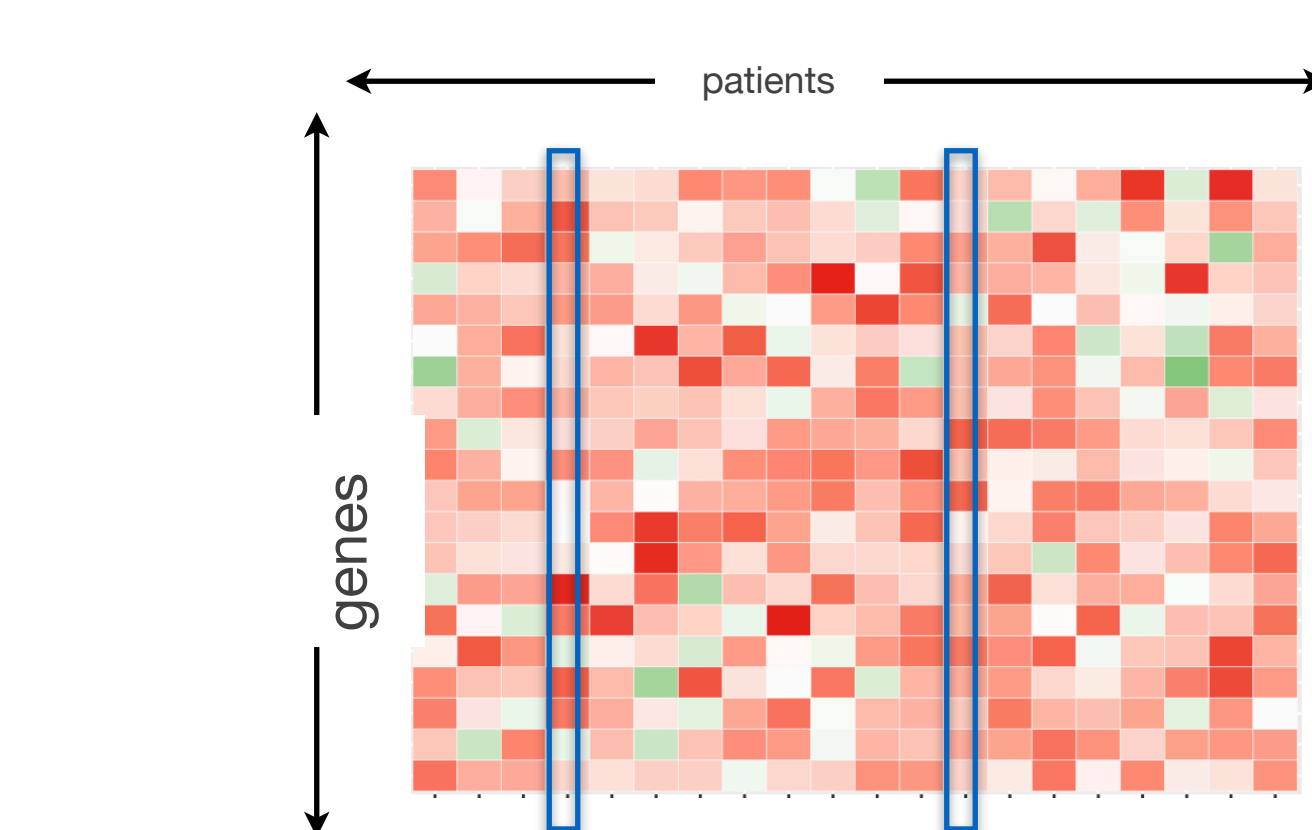


Finding structure in the data

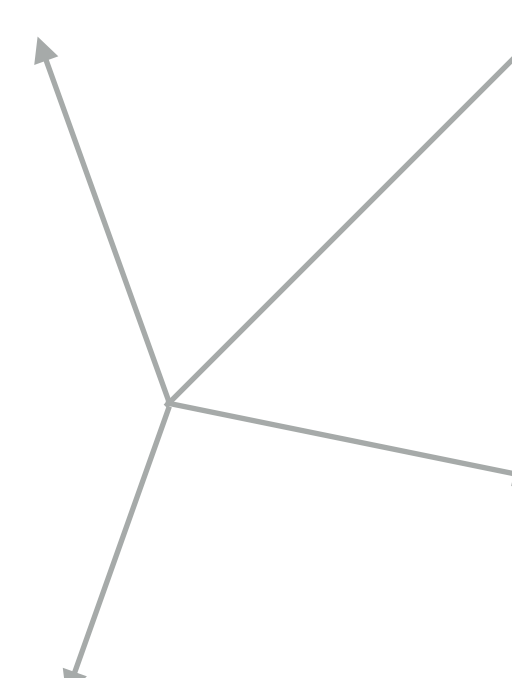
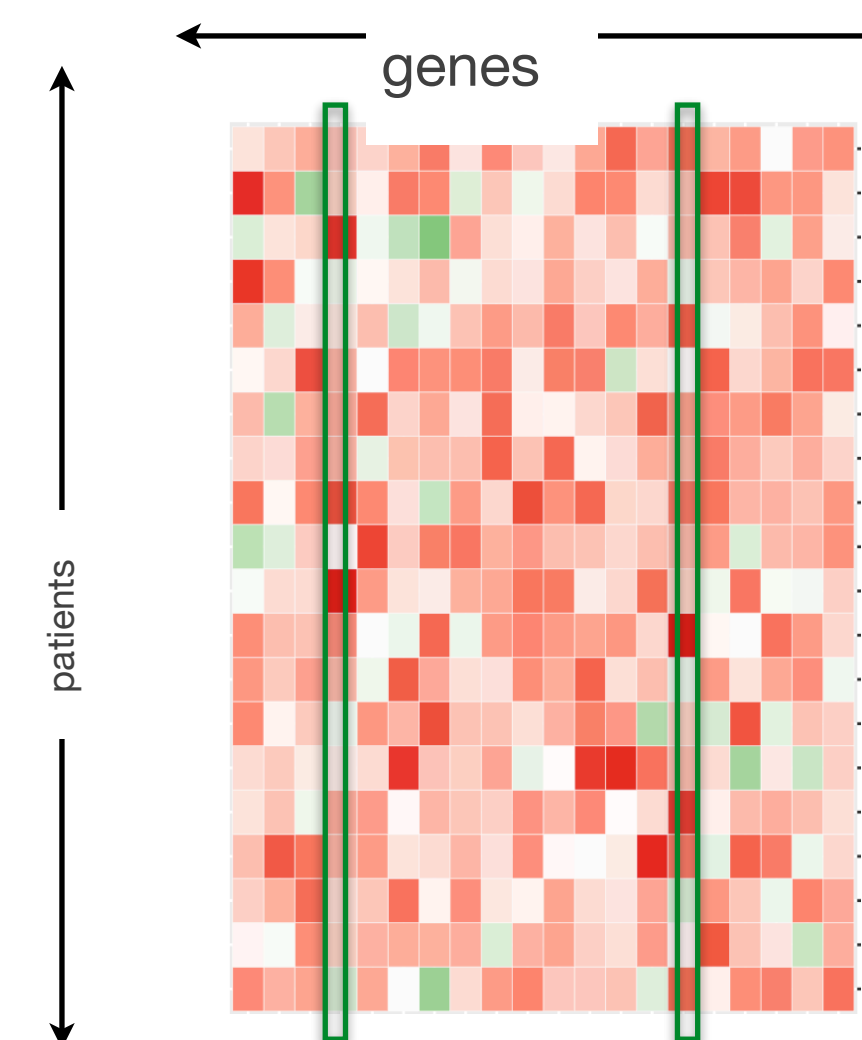
- So far, we have looked at individual variables (distribution, mean, spread,...) and the relationship between **variables**
- We now want to find a structure between **samples** (groups!)

- group similar samples
→ **clustering**
- show variables with high variance
→ **principal component analysis**





Points = **patients** in
20,000 dimensional
gene space

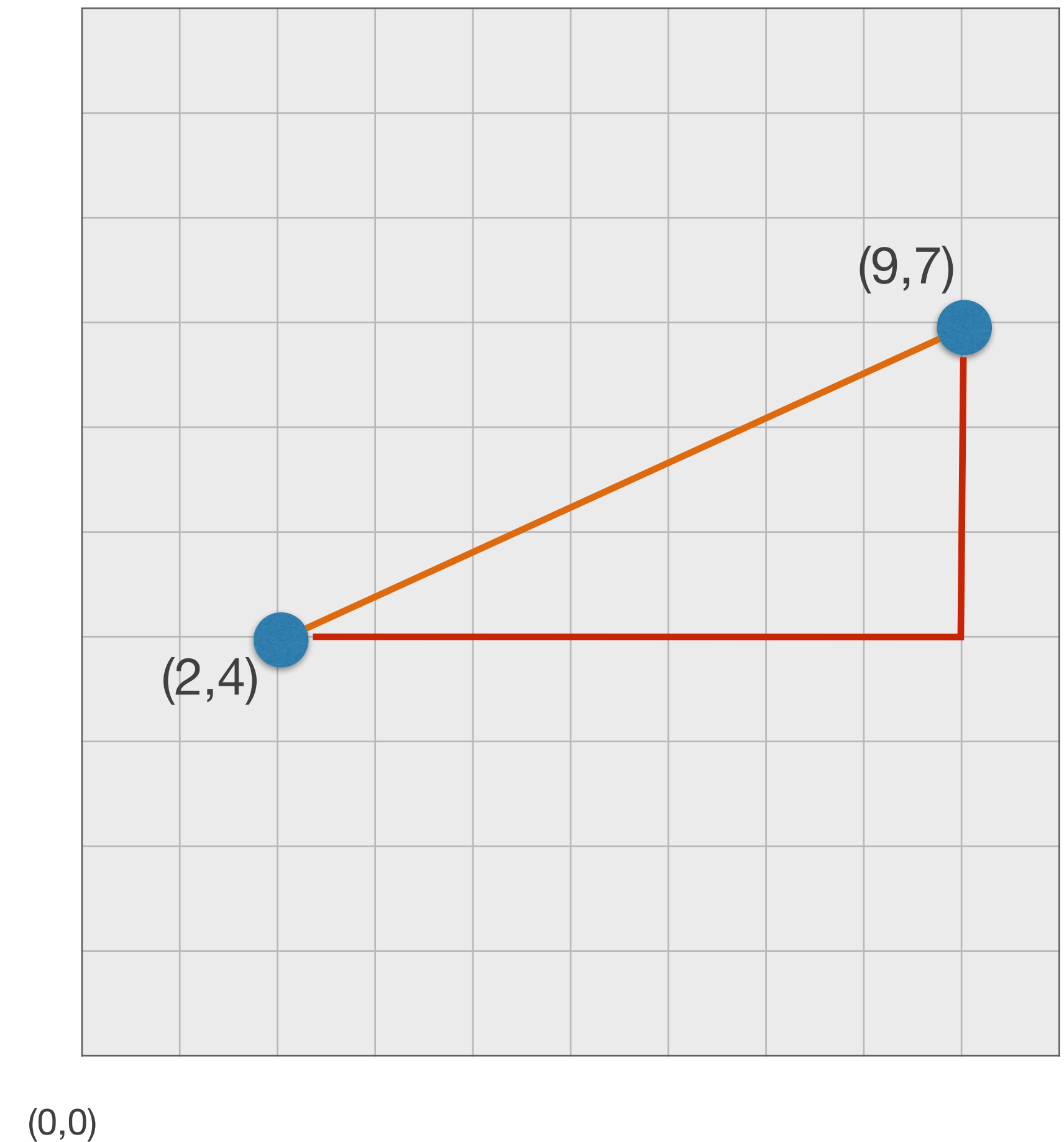


Points = **genes** in
500 dimensional
patient space

Distances / dissimilarities

- Distances should be
 - positively defined : $d(a,b) \geq 0$
 - $d(a,a) = 0$
 - $d(a,c) \leq d(a,b) + d(b,c)$
- Several distances can be defined in n-dimensional space
 - Euclidean distance**
 $d^2 = (9-2)^2 + (7-4)^2 = 58$
 - Manhattan distance**
 $d = 7 + 3 = 10$
 - Correlation distance** (when dimension > 2)

$$d = \frac{1}{2}(1 - r) \quad r = \sum_{i=1,2} \frac{(x_i - \bar{x})(y_i - \bar{y})}{\sigma_x \sigma_y}$$



Distances / dissimilarities in any dimensions

- Euclidean distance

$$d_{Euclidean}(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

- Manhattan distance

$$d_{Manhattan}(x, y) = \sum_{i=1}^n |x_i - y_i|$$

- Correlation distance

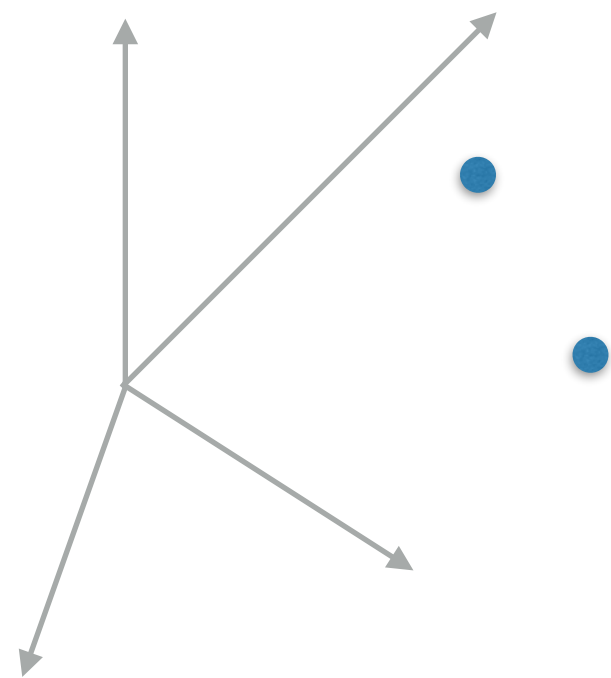
$$r = \frac{1}{n-1} \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{\sigma_x \sigma_y} \quad d_{corr}(x, y) = \frac{1}{2}(1 - r)$$

- Maximum distance

$$d_{Max}(x, y) = \max |x_i - y_i|$$

Example

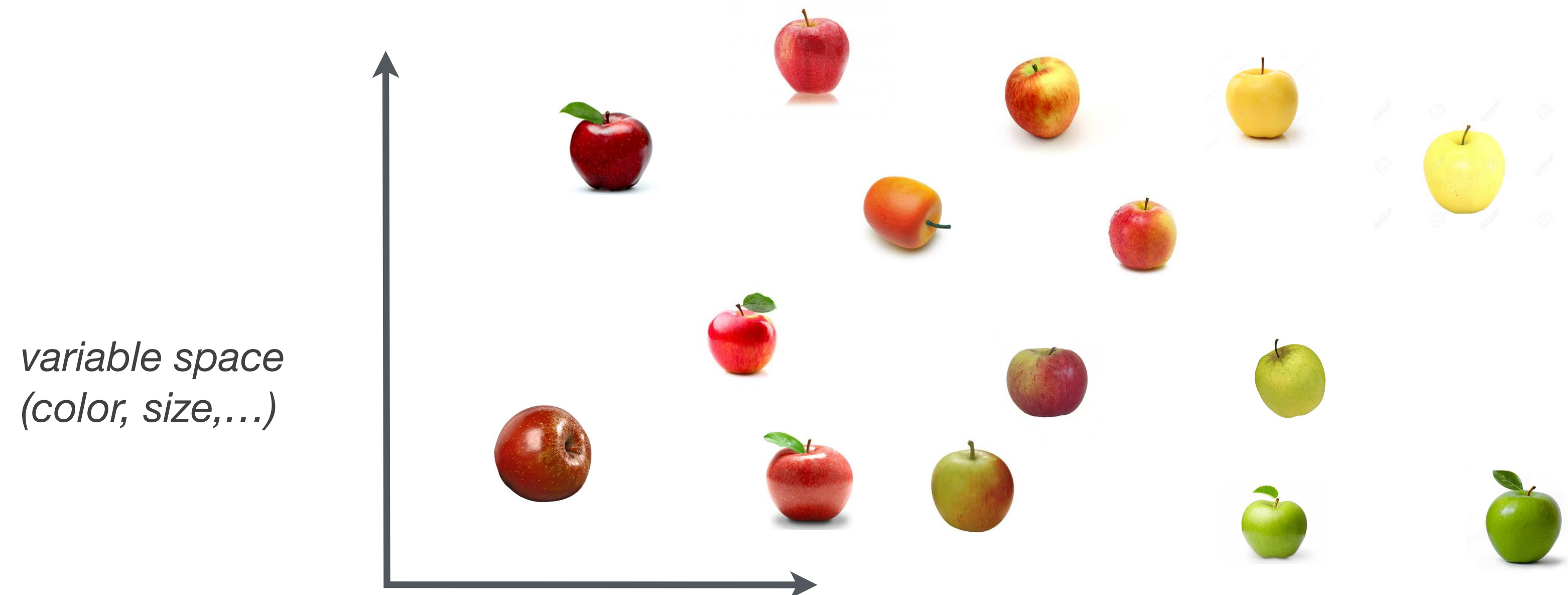
	variable 1	variable 2	variable 3	variable 4
x	-0.7152422	-0.7526890	-0.9385387	-1.0525133
y	-0.4371595	0.3311792	-2.0142105	0.2119804



Distance (x,y)	Value
euclidean	2.00
manhattan	3.70
maximum	1.26
correlation distance	$0.5(1-0.16) = 0.42$

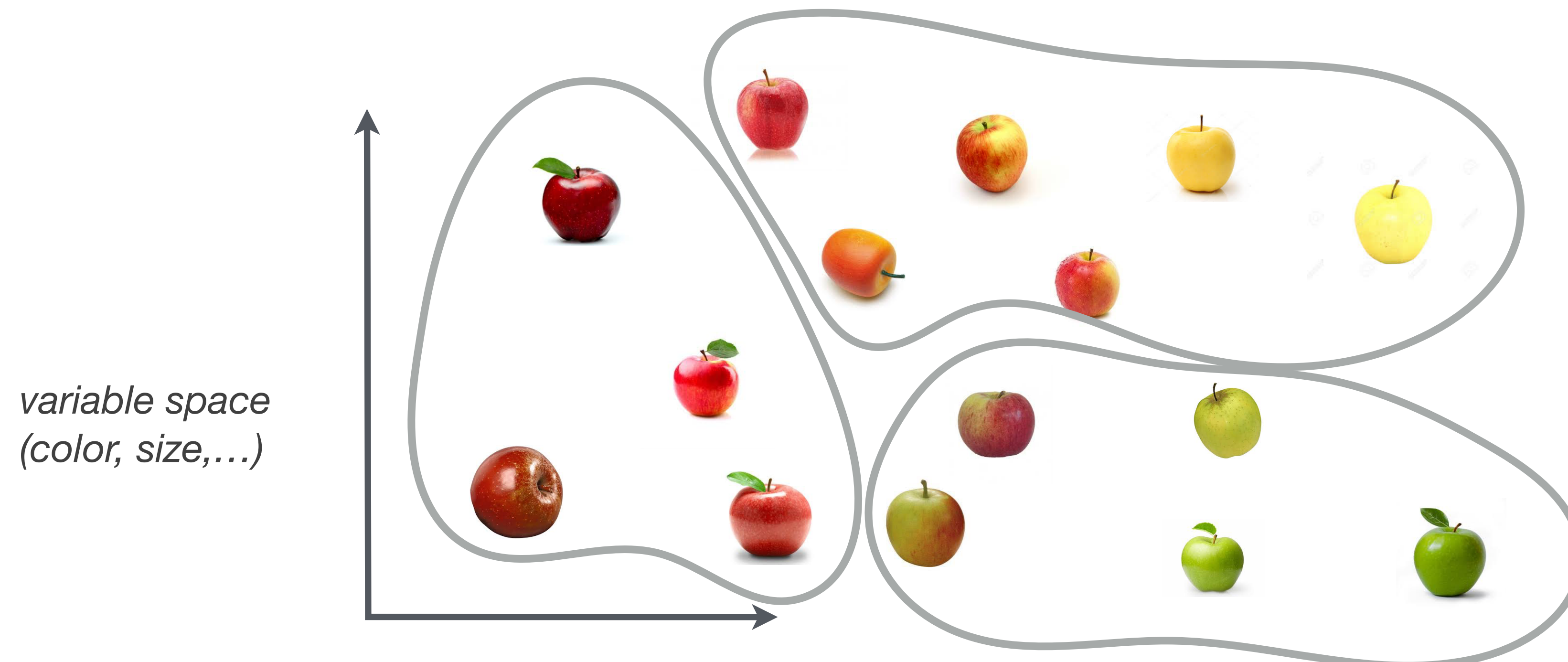
Clustering data

- Idea: group objects (= samples) which are **close to each other** in the variable space
- this implies the definition of a **distance between objects**
- **unsupervised approach:** we do **NOT** know to which group which sample belongs!



Flat clustering

- **Partition samples** into k -groups
- Important questions
 - how to define a **distance**?
 - optimal **number k of groups**?
 - how to deal with **ambiguous** samples?



4. Finding structure in the data - k-means clustering

k-means clustering

- **Objects** in a d-dimensional space
 - points in a 2d euclidean space
 - patients in a 20000d space of gene expression, ...
- Distance between object is the **euclidean metric**
- Define an **expected number k of clusters**
 - try out different numbers k
 - use external knowledge
- See demo here
<https://user.ceng.metu.edu.tr/~akifakkus/courses/ceng574/k-means/>

k-means



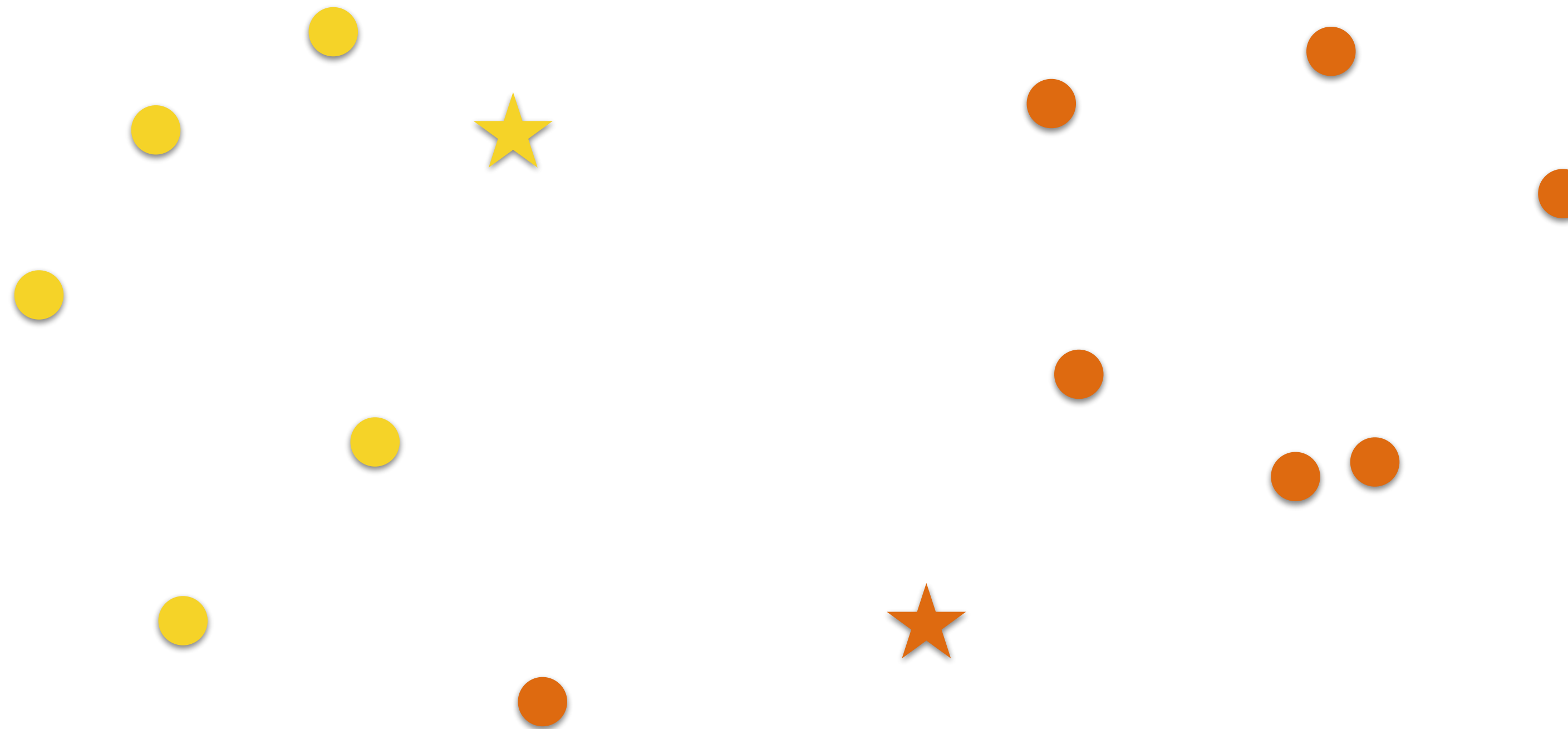
k-means

- **Step 1:** define k random centers (here: $k=2$)
- assign each point to the closest one of these centers



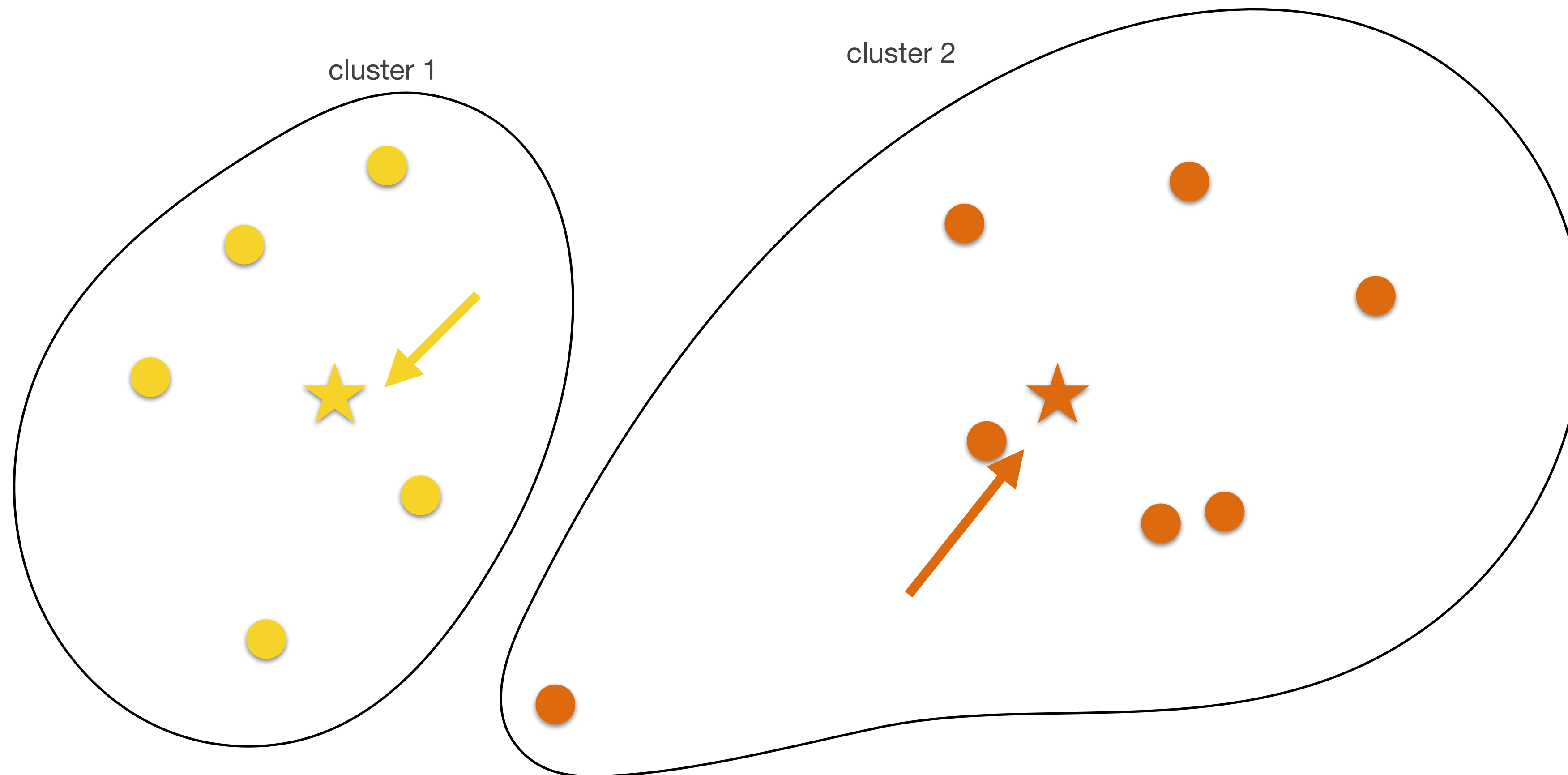
k-means

- **Step 1:** define k random centers (here: $k=2$)
- assign each point to the closest one of these centers



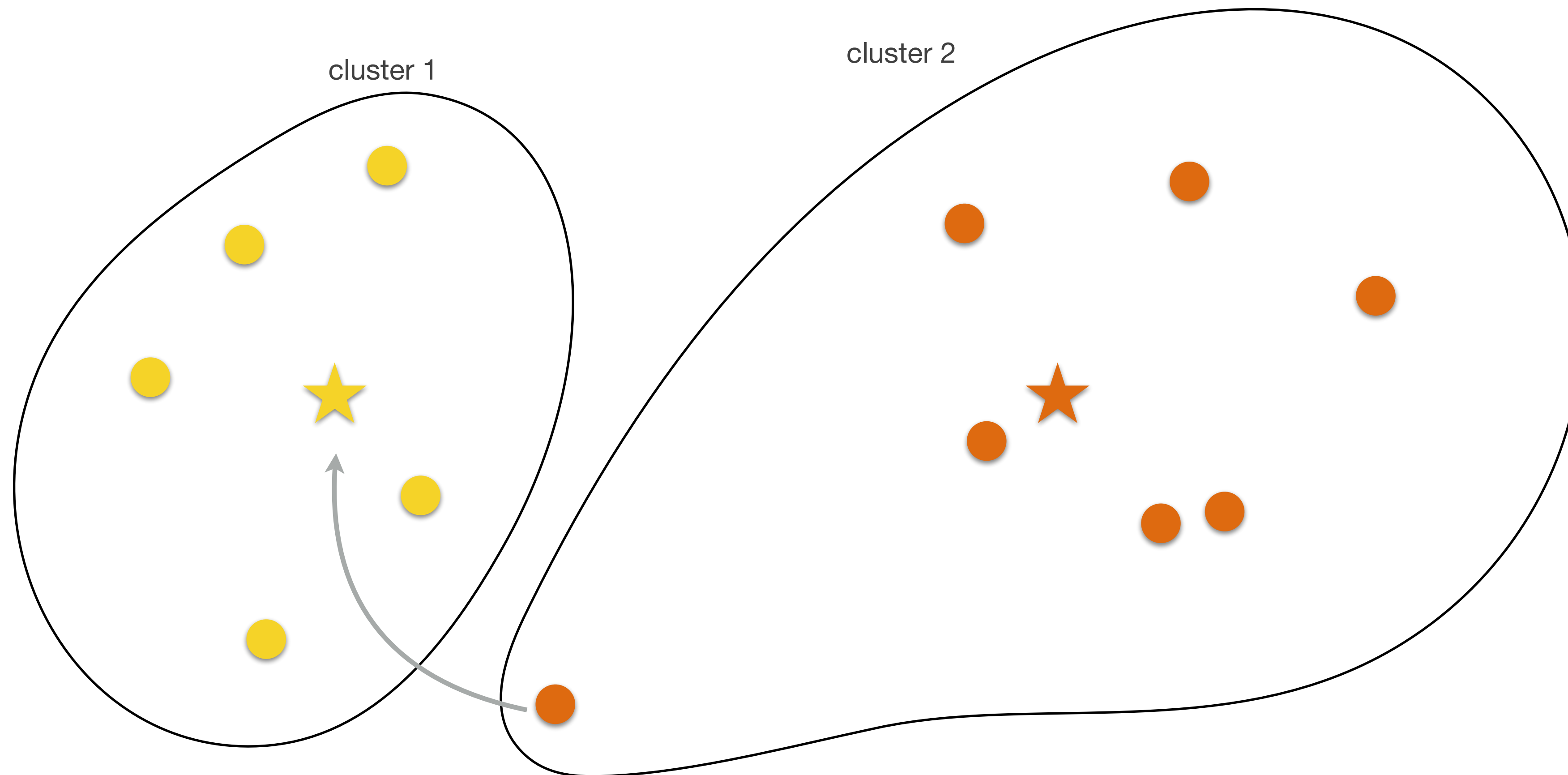
k-means

- **Step 2:** determine the **center of gravity** of these 2 groups of objects
- assign each point to the closest one of these centers



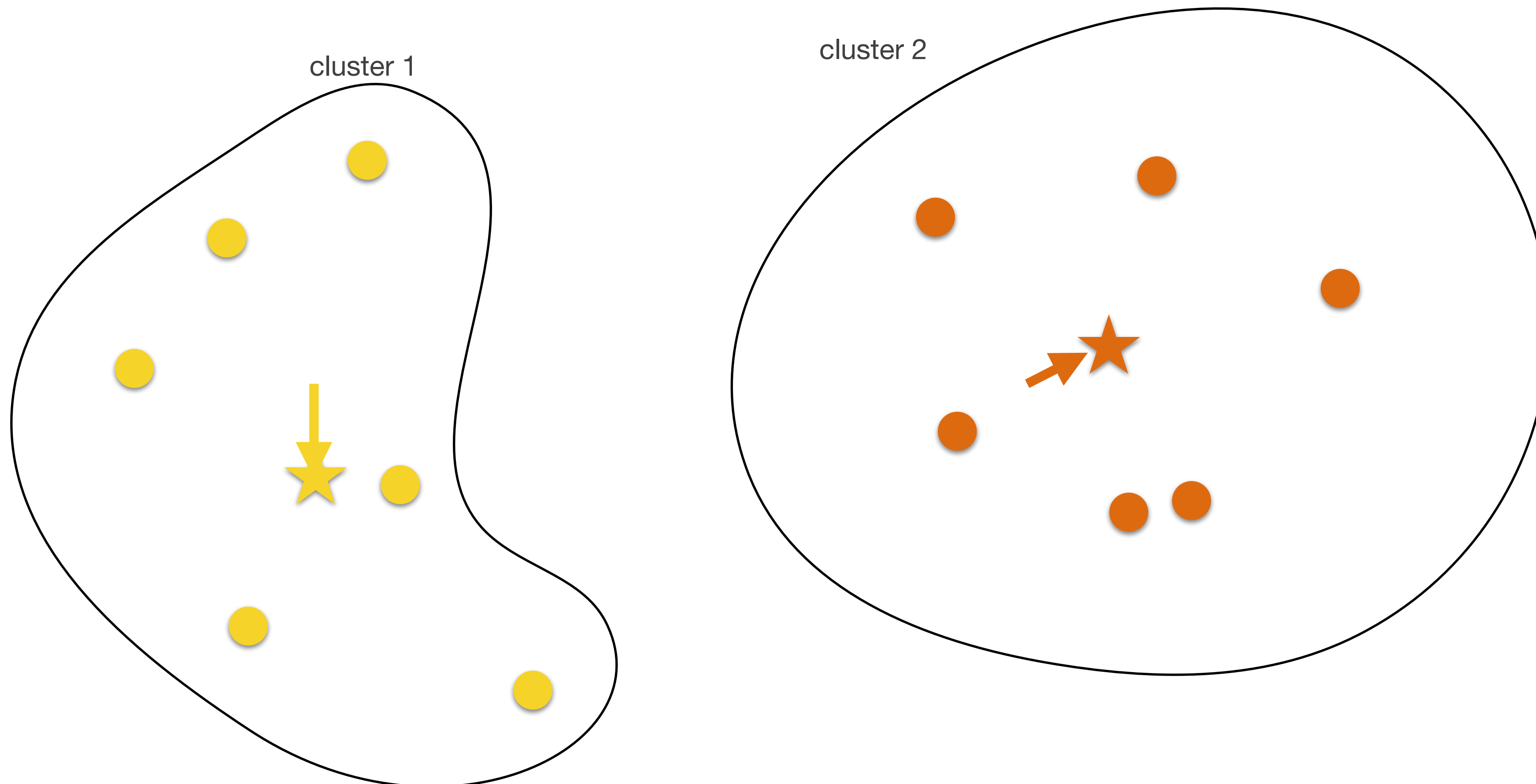
k-means

- **Step 3:** assign again each object to the updated centers
- assign each point to the closest one of these centers



k-means

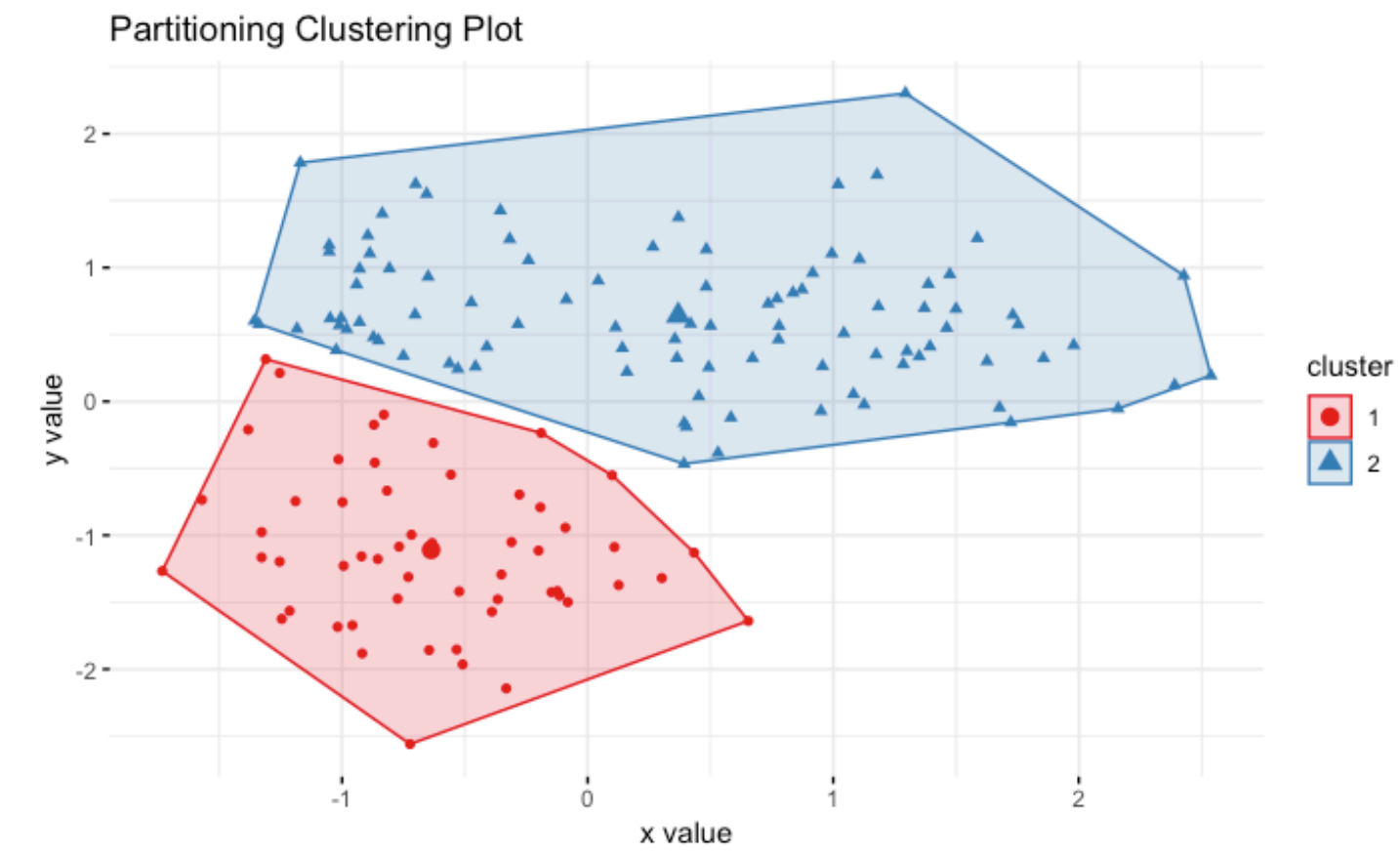
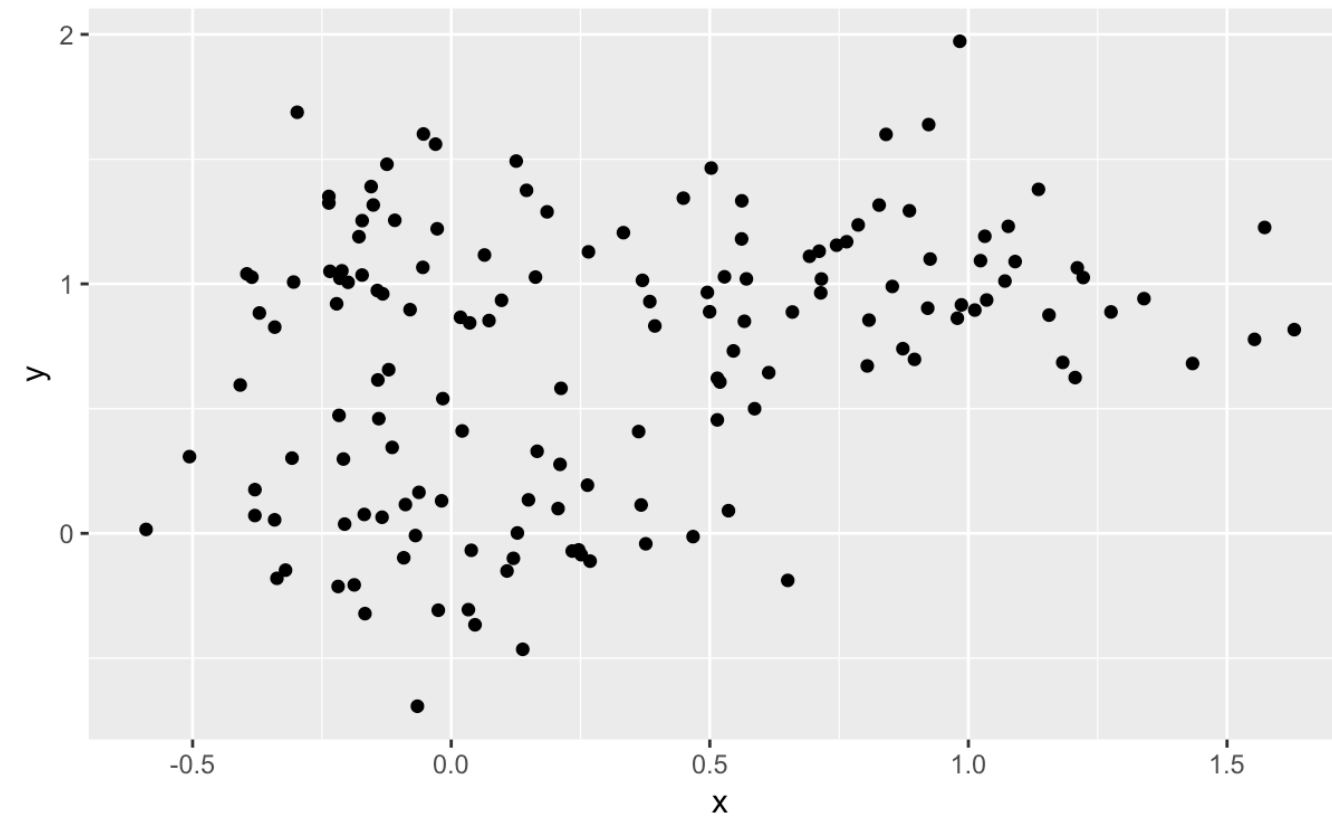
- **Step 4:** update the centers according to the new assignment, and start over



k-means

- When to stop:
 - when no object changes its cluster assignment anymore
 - or: when a certain number of max iterations are reached
- Process should be repeated over a number of random initial conditions!
- Verify the consistency of the clustering obtained using several random conditions
 - identical = very strong cluster structure
 - fluctuating = no clear cluster structure
- **Clusters will be identified, even if there is no group structure in the data!**

k-means - choosing k



Which of these clusterings is better and why?

k-means - choosing k

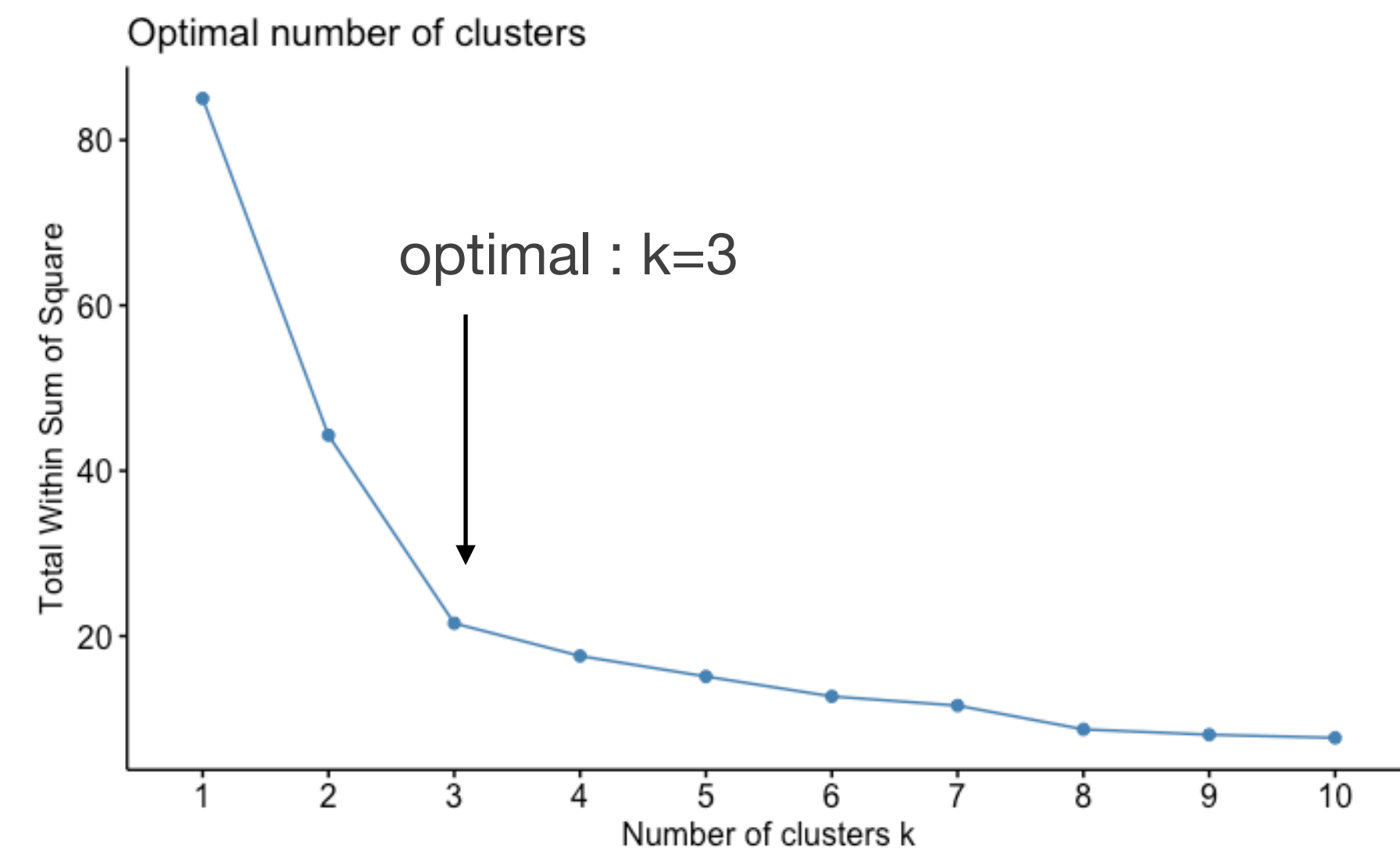
- **Fundamental idea**
in a good clustering, the **distance within clusters** should be much smaller
than the **distance between clusters**
- Several methods implement this principle
 - **elbow** method:
 - **silhouette** method
 - (many other methods exist to define optimal cluster number...)

k-means - elbow method

- Sum all **pairwise distances squared** between members of the same cluster (within square distance = WSS)

$$WSS = \sum_{i,j \in \text{same cluster}} |x_i - x_j|^2$$

- If each point is its own cluster, then WSS = 0
- WSS decreases with increasing k
- when optimal number of clusters is reached, adding additional clusters does not improve the WSS that much
→ kink (or “elbow”) in the curve



k-means - silhouette method

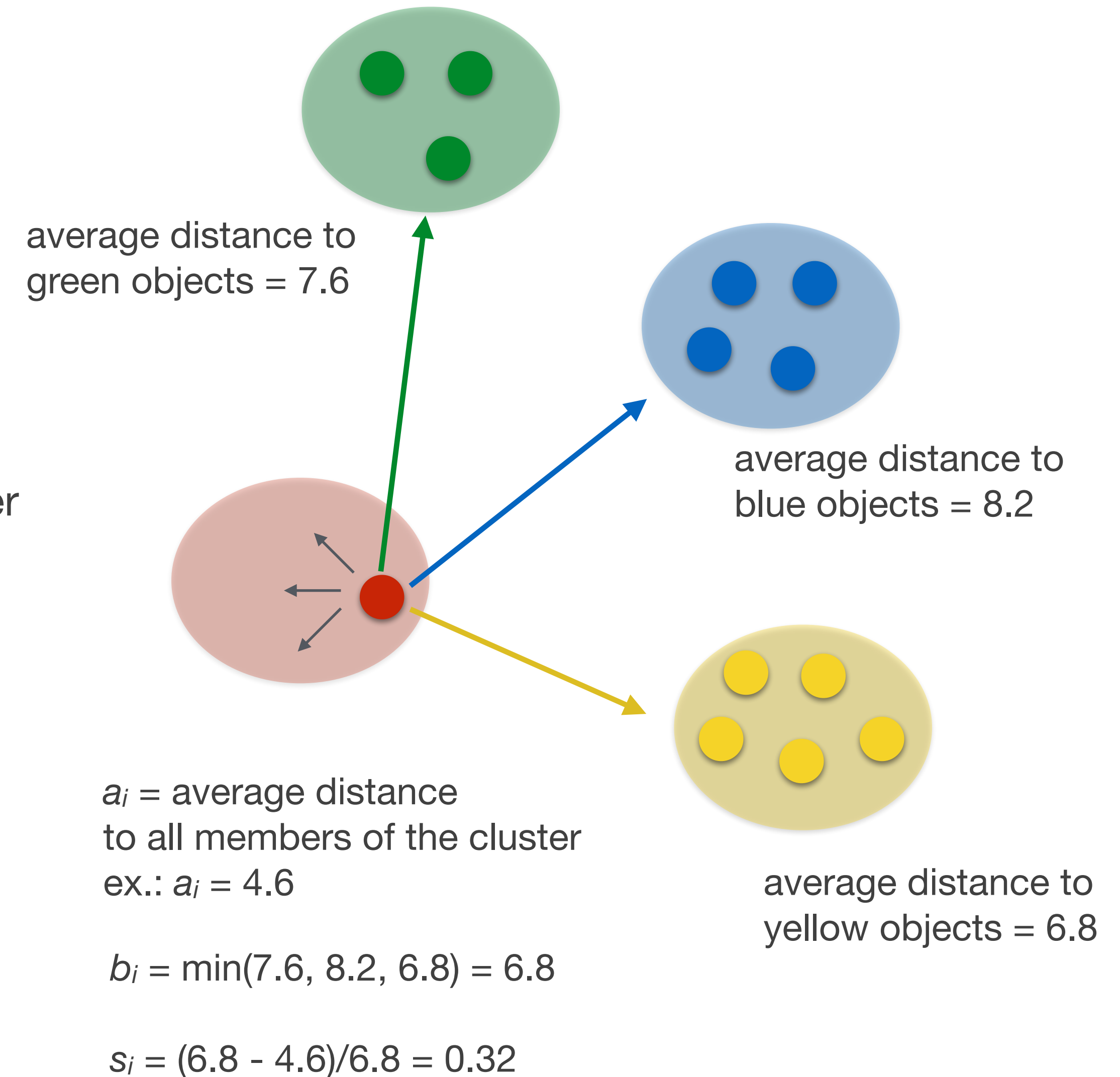
- for each element i , compute
 - mean distance a_i to all members of his cluster
 - smallest average distance b_i to members of all other clusters

- Define silhouette value as

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)} \quad s_i = 0 \text{ if only one point in cluster}$$

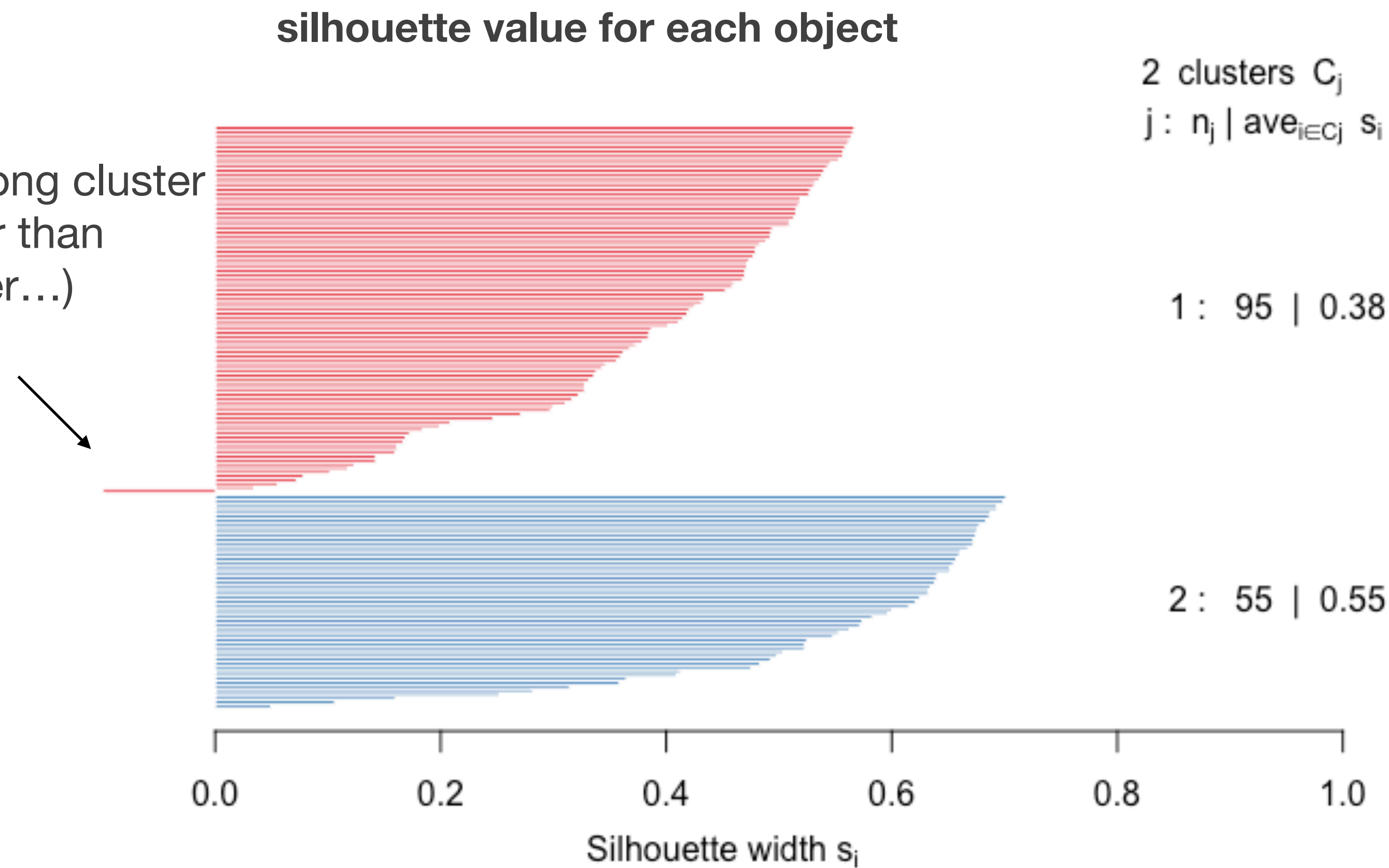
- Properties

- $s_i \sim 1$: object very well clustered
- $s_i \sim 0$: ambiguous object
- $s_i < 0$: wrongly assigned
- Convention : $s_i = 0$ if only one element in the cluster!

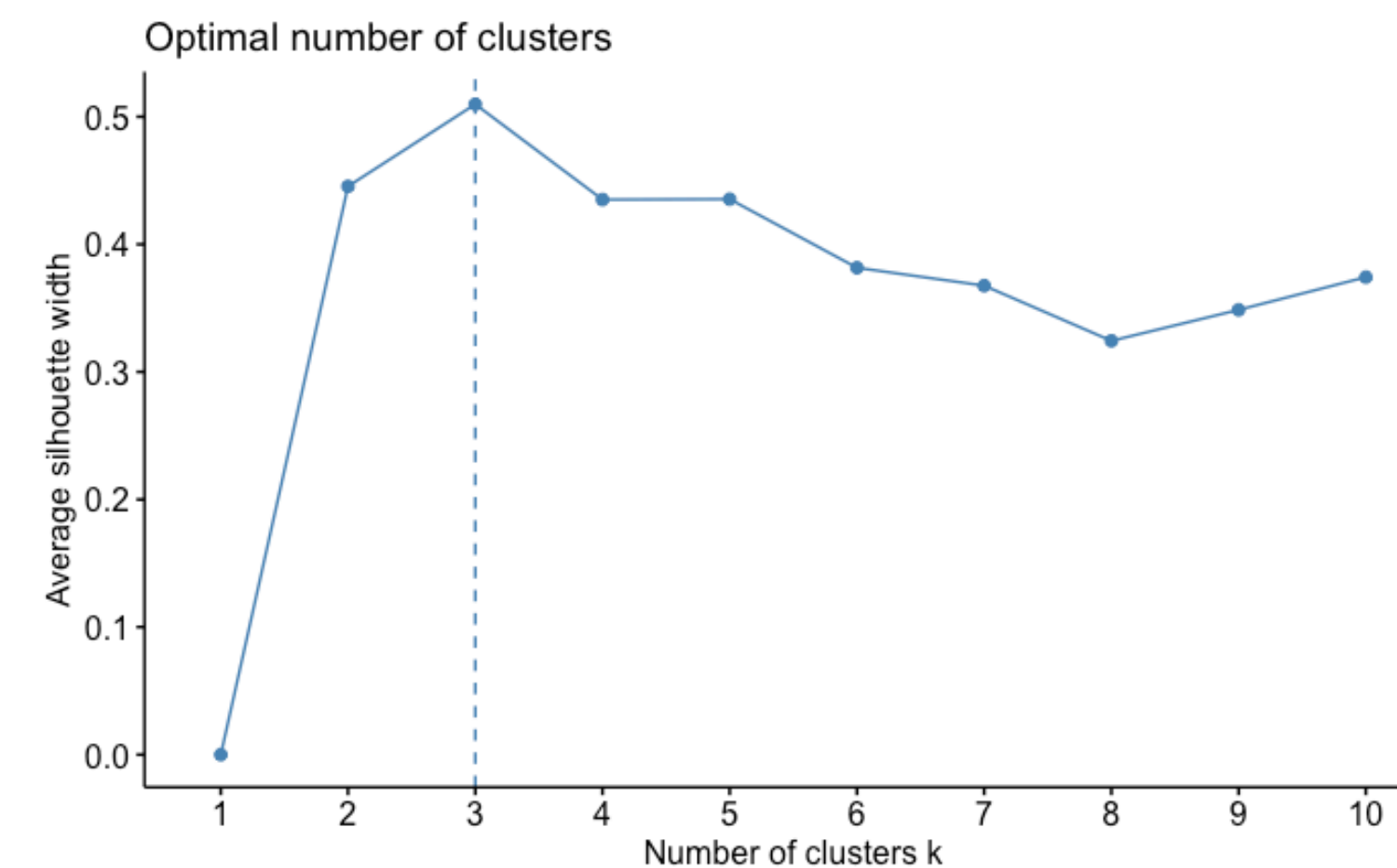
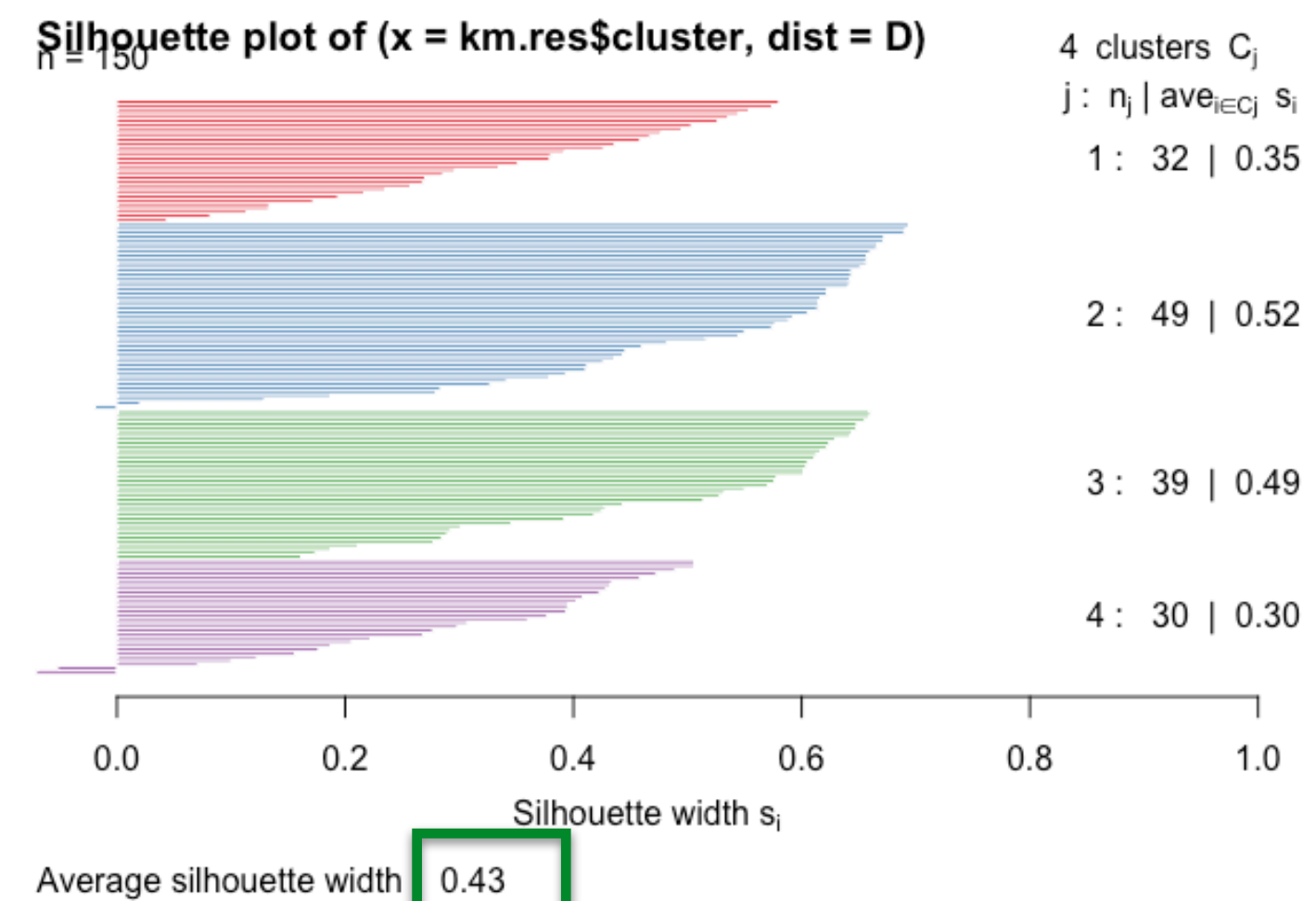
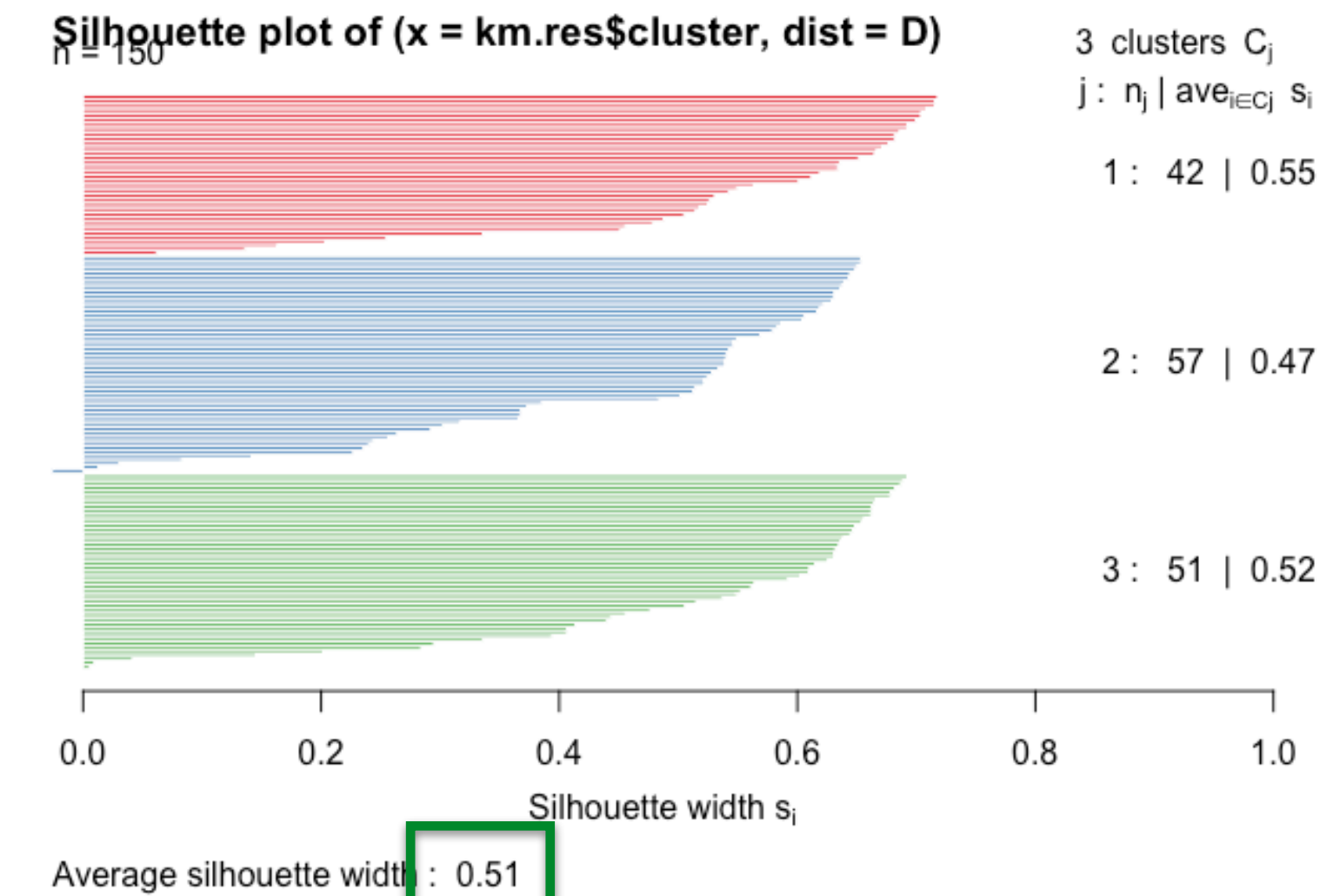
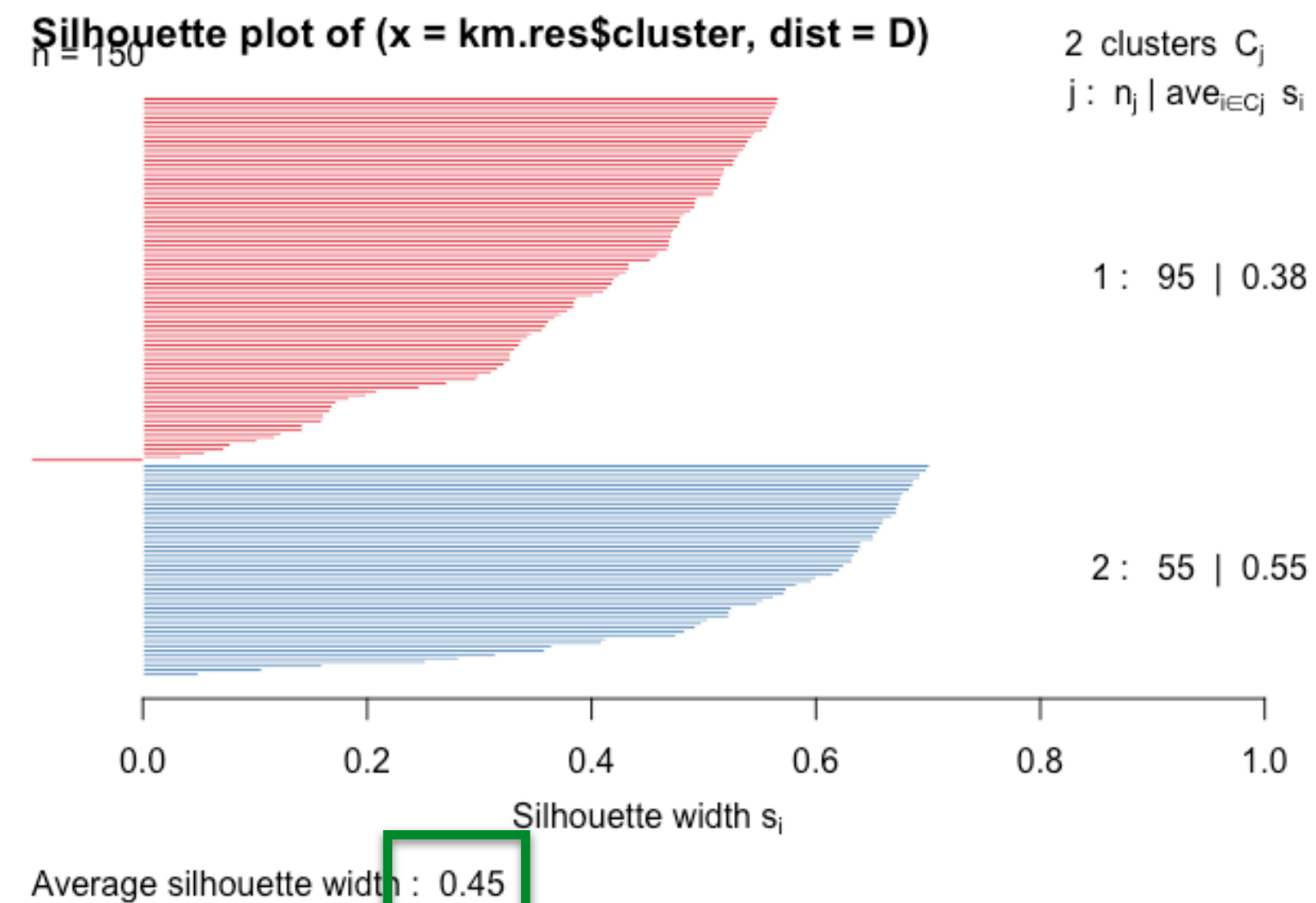


k-means - silhouette method

negative values:
object is assigned to the wrong cluster
(closer to some other cluster than
to the members of his cluster...)



k-means - silhouette method



Week 2 - stop and think

*First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!*

1. What are the correct statements about clustering
 - A. clustering is a supervised process, i.e. you need labels for the data points
 - B. increasing the number of clusters makes the clustering better
 - C. a high within-cluster and low between cluster distance means bad clustering
 - D. The choice of distance or similarity measure can change the resulting cluster structure.
2. Is it theoretically possible to find an example in which Pearson and Spearman correlations have opposite signs? Think before asking your favorite AI...
3. Is Pearson always higher/lower than spearman? Find examples...
4. If k-means can in principle give you different results depending on the random initialization, how would you decide which is the best?
5. For $k=2$, imagine a distribution of points in 2d for which k-means would give very different clusterings depending on the initialization...
6. For $k=2$, imagine a configuration of 10 points such that the silhouette score for one of the points would be negative.